

Project Report Template

Advanced Analytics and Applications

Team 07



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2026-04-01

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1 Introduction

1.1 Motivation and Problem Statement

The ride-hailing market is already crowded. Uber, Lyft, FreeNow, Lime, and Tier compete for the same trips, and current market reports point to ride-hailing, increasingly paired with autonomous and electric vehicles, as the segment that will dominate urban mobility in the near term. Our client, a well-established German car manufacturer, is building a ride-hailing platform on a fully electrified vehicle fleet, wanting to launch first in the United States, where customers have proven more receptive to novel mobility business models.

The client can design and build vehicles, but has no experience running a ride-hailing fleet. Managing an electric fleet adds a further constraint: Charging schedules directly affect vehicle availability in ways a conventional fleet does not face. To plan routes, size a fleet, and decide where to deploy vehicles, the client needs a clear picture of how demand moves through a city across space and time.

This report builds from historical taxi trip data for Chicago, made available by the Chicago Data Portal and covering trips from 2024 onward. Taxis are not ride-hailing vehicles, but in a market the client has not yet entered, they are the closest available proxy for one: metered urban trips, dispatched on demand, subject to the same weather and time-of-day patterns that will govern the client's own fleet. We supplement the trip records with independently collected weather data and, where it adds value, point-of-interest data from OpenStreetMap, and we analyze demand at spatial resolutions ranging from coarse census tracts to fine-grained H3 hexagon grids, and at multiple temporal resolutions, to establish which level of detail the data can actually support. We also investigate how a vehicle should charge at a driver's home between shifts, given a battery of finite capacity and a next-shift energy requirement that is not known with certainty in advance, using Reinforcement Learning.

1.2 Research Questions

The analysis that follows is organized around five questions, each feeding into the next.

1. How does taxi demand vary across space and time in Chicago, and what plausibly explains the patterns observed at different spatial units and temporal bin sizes?
2. Where are the persistent spatial hot spots of demand, and what do Gaussian mixture models and kernel density estimation reveal about their structure?
3. Can demand be predicted for a specific spatial unit and time bucket rather than simply extrapolated from the previous hour, and how does a support vector machine compare against a feedforward neural network on this task?
4. What charging policy should an electric taxi driver follow at home, given a charging rate that can be adjusted every fifteen minutes, a battery capacity that cannot be exceeded, and a next-shift energy requirement that is only known probabilistically at the moment the driver departs?
5. What do these findings imply for the client's fleet operations, and in particular, should the client invest in private home charging infrastructure or rely on public charging networks instead?

2 Data Description

2.1 Data Sources

Three raw datasets underpin the analysis: taxi trip records, hourly weather observations, points of interest, accompanied by a boundary layer of Chicago’s community areas. Each is described below by origin, coverage, and native granularity; the cleaning and feature-engineering steps applied to each follow in Section 3.

The taxi trip records come from the Chicago Data Portal’s Taxi Trips dataset, which publishes one row per dispatched, metered taxi ride in the city. The raw download spans January 1, 2024 through May 31, 2026 and holds 16,086,255 trips across 23 fields: trip and taxi identifiers, start and end timestamps, duration and distance, the fare broken into its components and as a total, the operating company, and pickup and dropoff locations given as census tract, community area, and centroid coordinates. Locations are masked for privacy, so a published coordinate is the centroid of the containing census tract rather than the true pickup point, and the tract itself is withheld on just over half of all trips (55.3% of pickups, 56.6% of dropoffs) against 2.8% and 8.8% for community area. Each trip is a ride requested wherever and whenever a passenger wanted one, which is the demand pattern the client’s own fleet will have to anticipate.

Weather observations come from the ERA5-Land reanalysis product, pulled through the Copernicus Climate Data Store API for a single coordinate near Chicago (latitude 42, longitude -87.8) rather than a network of ground stations. The record runs from January 1, 2024 through May 12, 2026 at hourly resolution and carries six variables: 2-metre air temperature, total precipitation, snow cover, snow depth, and the two horizontal components of wind at 10 metres. A reanalysis is a modeled reconstruction rather than a direct station reading, but it gives complete, gap-free hourly coverage for the entire trip period, which no single Chicago weather station could promise.

Points of interest come from OpenStreetMap, extracted with OSMnx for the “Chicago, Illinois, USA” boundary and limited to three tag families: amenity, shop, and tourism. The raw download returns 43,237 points. Unlike the trip and weather data, this is a snapshot of the map as it stands today, not a historical record tied to the trip period. The boundary layer for Chicago’s 77 official community areas is used for spatial reference and choropleth mapping throughout the report. It is a distinct unit from the census tracts already present in the trip data itself.

3 Data Preparation

This section describes the preparation pipeline that turns the raw sources described in Section 2 into the analysis-ready datasets used throughout the report.

3.1 Taxi Trip Description and Cleaning

Cleaning the raw trip table (Section 2) proceeds in three stages: trimming columns, removing unusable rows, and splitting the result into the two parallel datasets the rest of the pipeline consumes.

Seven of the 23 raw fields are dropped as redundant. The individual fare components (fare, tips, tolls, and extras) are already summed into the trip total, the payment type says nothing about where or when demand appears, and the two point-geometry columns merely repeat the latitude and longitude stored separately. Sixteen fields remain: trip and taxi identifiers, start and end timestamps, duration and distance, pickup and dropoff census tract and community

area codes, the fare total, the operating company, and pickup and dropoff coordinates. The export writes decimals with commas and prefixes currency with a dollar sign, so these columns are parsed locale-aware before any numeric comparison is made on them. From the coordinates we also derive an H3 cell index per trip at resolutions 7, 8, and 9, which gives the hexagonal units used alongside tracts and community areas later.

Rows are removed on two grounds. Records missing a taxi identifier, an end timestamp, a duration, a distance, or a fare total are dropped outright; each of these is absent on well under 0.2% of trips, a share small enough that imputation would add assumptions without adding information. A sanity filter then removes trips with a non-positive duration or distance, or a negative total, none of which describe a journey that actually happened. No statistical outlier removal is applied beyond that: extreme but plausible values, such as long airport transfers, are legitimate demand and are deliberately kept.

Because census tract is withheld far more often than community area, the cleaned data is kept as two parallel datasets rather than forcing one compromise on every downstream analysis. The larger retains every trip with a known community area; the stricter keeps only those that also carry a census tract. Both are truncated to the last hour the weather series covers, so that no model can later be trained or evaluated against forward-filled weather. That leaves **12,949,755 trips at community-area completeness** and **6,052,512 at census-tract completeness**, both spanning January 1, 2024 through May 12, 2026.

3.2 Weather Data

3.3 Points of Interest

Point-of-interest data was optional under the assignment brief, but there is precedent for treating it as a demand signal: Yan et al. (2020) predict taxi demand at city scale by combining several sources of urban geospatial data, point-of-interest records among them. We therefore include POI density to test how much it explains about where and when ride demand originates in Chicago. Coverage across the three tag families is uneven: 7,053 records lack an amenity tag, 37,552 lack a shop tag, and 41,738 lack a tourism tag, so most points carry only one of the three. We collapsed them into a single `poi_type` field by taking amenity where present and falling back to shop, then tourism, which left every point with one of 311 distinct raw types and none without a type or a geometry.

At 311 types the data is too granular to serve directly as model features, so we mapped each raw type onto one of fourteen broader categories that plausibly drive ride-hailing demand: food and drink, nightlife, entertainment, leisure and sports, grocery, shopping, health, finance, education, lodging, transport, automotive, services, and civic and community. The mapping doubles as a filter. Fifty-five raw types like infrastructure and street furniture (benches, parking entrances, post boxes, waste bins), fit no category and were dropped, leaving 20,108 categorised points of interest. Roughly half of those are building polygons rather than single points, so each was reduced to its centroid and assigned to a spatial unit, then aggregated to all three units used downstream: H3 resolution 7, census tract, and community area. The handful of points that fall just outside the official city boundary are excluded from the tract and community aggregations but retained at the H3 level, which is defined continuously over any coordinate.

3.4 Spatial and Temporal Discretisation

3.5 Feature Engineering

4 Data Analytics

This section covers all analytical methods applied to the prepared data: descriptive spatial analytics, predictive modelling of demand (SVM and neural networks), and reinforcement learning for smart charging.

4.1 Descriptive (Spatial) Analytics

To better understand the characteristics of taxi demand before developing predictive models, we first perform a descriptive analysis of the dataset. The analysis investigates spatial and temporal demand patterns, examines trip characteristics, compares different spatial aggregation levels, and identifies persistent demand hot spots. **### Demand Patterns** The taxi demand data exhibit strong spatial and temporal heterogeneity. Most trips are relatively short in terms of both travel distance and duration, while trip fares show a pronounced right-skewed distribution. Consequently, the majority of journeys are inexpensive, whereas a comparatively small number of long-distance trips account for substantially higher fares. Spatially, taxi activity is highly concentrated in a limited number of locations. Pickup and dropoff demand are strongly correlated, indicating that areas generating many departures also receive many arrivals. However, comparing pickup and dropoff distributions reveals localized differences that likely reflect commuting flows, airport traffic, and the separation of residential and commercial districts. Additional descriptive statistics and distribution plots are provided in **Appendix A**.

4.1.1 Spatial Patterns

Taxi demand was analysed using community areas, census tracts and H3 hexagonal grids at multiple resolutions. Although all representations reveal similar large-scale demand structures, the selected spatial resolution considerably influences the observed level of detail. Community areas provide a coarse overview of demand, while census tracts capture substantially finer local variation. H3 resolution 7 offers a compromise between spatial detail and robustness, whereas H3 resolution 8 largely reproduces the census tract representation. This behaviour results from the privacy-preserving coordinate masking applied to the original dataset, where published coordinates correspond to census tract centroids whenever available. **[Figure X: Spatial demand distribution]** Across all spatial resolutions, demand remains highly concentrated in only a small number of regions. Increasing spatial resolution improves the visibility of localized demand patterns but simultaneously introduces considerable sparsity, with many spatial units containing few or no observations during large parts of the observation period. Further spatial comparisons, concentration measures, Lorenz curves and pickup-dropoff difference maps are shown in **Appendix A**.

4.1.2 Temporal Patterns

Taxi demand follows clear and recurring temporal patterns. Demand is lowest during the early morning hours, increases throughout the day and reaches pronounced peaks during commuting and evening leisure periods. Distinct differences are also observed between weekdays and weekends, reflecting changes in travel behaviour. **[Figure X: Average hourly taxi demand]** Besides the intraday cycle, demand also varies systematically across weekdays and seasons, indicating that temporal information is highly informative for subsequent forecasting

models. Additional temporal analyses, including weekday-weekend comparisons and seasonal demand patterns, are provided in **Appendix A**.

4.1.3 Effect of Spatio-Temporal Resolution

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4.1.4 Demand Hot Spots

Persistent demand hot spots were identified using Gaussian Mixture Models (GMM) and Kernel Density Estimation (KDE). Both approaches consistently identify the central business district and major transportation hubs as the dominant centres of taxi activity. **[Figure X: Gaussian Mixture Model hot spots]** The probabilistic clustering produced by the GMM groups regions with similar spatial demand characteristics, while KDE provides a continuous representation of demand intensity. Both methods highlight areas associated with dense commercial activity, transportation infrastructure and entertainment facilities, demonstrating the strong influence of land use on taxi demand. The complementary KDE visualisation is presented in **Appendix A**.

4.1.5 Weather Effects

4.2 Predictive Analytics (SVM & Neural Networks)

Fleet planning needs to know where and when demand will appear before the shift starts, so we predict trip starts per spatial unit and time bucket from calendar, location, weather, and POI features alone, without extrapolating one hour from the previous one. The question the model answers is a scenario question: how much demand to expect on a rainy Sunday morning on the outskirts. We compare two model families on this task, a Support Vector Machine (fitted both directly on demand and on the residual over a historical-mean baseline) and a feedforward neural network, across six grids that cross three spatial units (H3 resolution 7, census tract, community area) with hourly and daily time bins. Both are scored against the same historical-mean baseline, which is what lets us ask whether a learned model, and a deep one in particular, is worth its added complexity here.

4.2.1 Problem Setup and Validation Strategy

The prediction target is `Total_Trip_Start`, the number of trips beginning in a given spatial unit during a given time bucket. Prediction is scenario-conditioned rather than autoregressive: no model sees the previous bucket's demand, only the cell's identity and location, the calendar, the weather, and the POI counts for that cell. Every model is trained on six grids: three spatial units (H3 resolution 7 with 108 active hexagons, 496 active census tracts, and 77 community areas) crossed with hourly and daily bins. Inactive units, those with no pickups anywhere in the observation window, are dropped during grid construction (Section 3). Sparsity differs sharply across the six: the share of zero-demand rows runs from 0.8% at community/daily to 96.5% at census/hourly. That spread drives the sampling decisions taken separately for each model family below. The reference point throughout is a historical mean per spatial unit, keyed by month and day of week on the daily grids and additionally by hour on the hourly ones. It is computed on the training split alone and merged into validation and test, so it carries no future information, and it serves twice over: as the `base_demand` feature available to both models and as the benchmark for the skill score, $1 - \text{MAE_model} / \text{MAE_baseline}$. Skill rather than raw

MAE is reported because mean demand differs by roughly a factor of 80 across the six grids, which makes absolute errors incomparable between them.

Train, validation and test are cut along the calendar rather than drawn at random. The earliest 50% of unique dates train the models (2024-01-01 $\rightarrow$ 2025-03-06), the next 20% serve for validation (2025-03-07 $\rightarrow$ 2025-08-26), and the final 30% are held back for testing (2025-08-27 $\rightarrow$ 2026-05-12). A random row-level split would leak autocorrelated demand into the test set, and it would also contaminate `base_demand`. Model selection is carried out on the validation split, with the test split scored once for the selected models.

4.2.2 Support Vector Machine

Two SVR strategies are trained on identical splits, samples and candidate feature sets, but differing in what they are asked to predict. The direct model regresses `log1p(Total_Trip_Start)` on the feature block and converts predictions back with `expm1`, clipped at zero. The residual model predicts the log-space correction to the historical mean instead, $(\log1p(\text{demand}) - \log1p(\text{base_demand}))$, and reconstructs counts as `expm1(log1p(base_demand) + correction_hat)`. It never sees the raw target, only how far a given cell and time bucket departs from its own historic mean. The motivation is that `base_demand` already carries much of the predictable signal, so a model asked to reproduce it from scratch spends its capacity on something the pipeline has already computed. Since that correction has a far smaller scale and variance than `log1p(demand)`, hyperparameters tuned for one target do not transfer to the other, and the residual model is therefore tuned from scratch.

SVR is inefficient to fit so each level is fitted on a class-balanced sample, for which we pull all non-zero demand rows from the training set and match it with at most an equal amount of zero demand rows so the target mix is 50/50 rather than the mix the level actually has (4000 rows for the kernel scan, 8000 for the grid search, 50000-row validation the level's winner re-scored on the full validation split). Levels with less zero-demand rows, the sample end up non-zero-heavy (community/daily with 0.8% zero). A uniform draw of the same size would be 94% empty cells at h3_7/hourly and 96.5% empty at census/hourly, and a model fitted on it would learn little beyond predicting zero. Validation and test are never rebalanced. Both are scored at natural prevalence, so the models are trained on a class mix they do not meet at evaluation.

Tuning proceeds in two stages per level and feature set. A kernel scan compares linear, polynomial and RBF at a fixed baseline setting, judged by validation R^2 in count space, and only the winning kernel carries into a three-fold grid search over `C` and `gamma`, extended by `degree` where the polynomial kernel wins, scored by count-space MAE. Four feature sets compete at every level (a basic calendar, location and history block, then that block plus weather, plus POI, and plus both). Table 1 lists every value searched and the sample sizes each stage is fitted on; Table 2 gives the feature blocks the four candidate sets are built from.

4.2.2.1 Validation Results

The direct model does not beat the historical mean it was given as a feature. At five of six levels its tuned winner is worse than the baseline on validation MAE, from -3.5% at community/daily down to -35.5% at census/hourly, and the single level it wins, h3_7/daily, it wins by only 0.2%. Predicting the correction instead improves the skill score at all six levels, but the improvement is uneven and mostly modest: community/daily is the one clear win at +5.3% ($R^2 = 0.970$), h3_7/daily turns its tie into a slim +1.2%, h3_7/hourly (+0.3%) and census/daily (+0.3%) remain ties, and community/hourly stays just short of the baseline at -1.0%. The exception is census/hourly, the sparsest grid at 96.5% zero rows, which loses heavily

under both strategies (-35.5% direct, -32.1% residual). The MAE skill is deeply negative while RMSE skill is roughly flat (+0.7% residual). Daily aggregation beats hourly at all three spatial resolutions under both strategies, consistent with hourly bins being sparser and noisier. Every winning direct model uses the flexible RBF kernel, while four of the six winning residual models do just as well with a straight-line fit (both h3_7 levels, census/hourly and community/daily). Both models work on a compressed scale and have their predictions expanded back into trip counts at the end, which means a small error before expansion becomes a large one after it. The direct model has to predict demand itself, anywhere from zero to several thousand trips a day, and needs the extra curvature to keep those errors from blowing up. The residual model only predicts a small adjustment to the historical average, so even a straight line stays close enough that the expansion cannot run away with it. All six residual winners also settled at $C=1$, against direct winners spread from $C=1$ to $C=50$.

The two best model are both residual models. At community/daily, fitted on the basic+weather feature set with a straight-line kernel at $C=1$, which brings the average error down from the baseline's 36.19 trips per area-day to 34.25 at an R^2 of 0.970. Second is the residual SVR at h3_7/daily, on the basic+poi feature set, also straight-line at $C=1$, moving 15.83 trips per hexagon-day to 15.64 at an R^2 of 0.954. Both sit on daily grids rather than hourly ones, both correct a historical mean instead of predicting demand outright, and neither needs a curved kernel. In operational terms the margin is small: roughly two trips per area per day at the better of the two, and about a fifth of a trip per hexagon at the other.

4.2.3 Deep Learning Model

For the neural network, we used a feedforward network that predicts log1p-transformed trip counts, which are then back-transformed via `expm1` for evaluation. We trained on the full training split without any zero/non-zero balancing, since our experiments showed that balanced sampling actually hurt the FNN's performance. For all six levels we used the same feature set, combining basic temporal and location features with all 14 POI categories and 5 weather variables. We did not test different feature set combinations as we did for the SVM, mostly due to computational constraints. The network uses ReLU activations in the hidden layers, a linear output unit, MSE as the loss function, and AdamW with a weight decay of $1e-4$. Training ran for up to 15 epochs with a batch size of 250. Within each training run, we used early stopping with a patience of 3 to select the best checkpoint based on validation loss, preventing the model from training past its best point. To find the best architecture per level, we ran a grid search over four layer configurations ([64,32,16], [128,64,32], [128,64,32,16], [256,128,64,32]), three learning rates (0.01, 0.001, 0.0001), and two dropout rates (0.0, 0.2), giving 24 configurations per level. The winning configuration per level was selected by validation MAE in count space. The final performance of each winning model was then evaluated once on the held-out test set, with results reported in (Figure X) across MAE, RMSE and R^2 . Looking at these results, h3_7/hourly comes out as the best performing level overall. As shown in (Table X), smaller and shallower networks worked best for the denser levels like h3_7/hourly and community, while census levels needed deeper networks and lower learning rates. Census/daily was the only level where dropout helped, which fits with it being the hardest level to train given its very high share of zero-demand rows. (Table X) shows the FNN results across all six spatial-temporal levels. The model beats the historical-mean baseline at four of the six levels, with h3_7/hourly performing best at a skill score of +6.3% MAE. The gains are small across all levels. Two levels perform notably worse: census/hourly is essentially a coin flip at -0.1%, and census/daily loses to the baseline by a large margin (-20.3% MAE, -44.7% RMSE). These are the two sparsest and most zero-inflated levels in our dataset, and the SVM struggles there as well, which suggests the difficulty lies in the data rather than the model choice. Overall, the FNN performs best at finer temporal resolution and

denser spatial units, while coarser and sparser levels remain hard to predict beyond a simple historical average. (Check after training)

4.2.3.1 Validation Results

4.2.4 Model Comparison

Both families are compared on the same held-out test split, at H3 resolution 7 and at both temporal bins, against the same historical-mean baseline. Census and community levels are excluded: the SVM was scored there, but no matching neural network prediction file exists, so a comparison at those levels would not be like-for-like. The comparison runs along six lines, each asking a different question about where a model's error actually falls. Supporting figures and tables are collected in Section A.

- **Aggregate accuracy.** MAE, RMSE and R^2 for the baseline, the SVM and the network at each temporal resolution, establishing whether either model beats the historical mean at all and by how much (Table 6).
- **Predicted against actual.** Predicted versus observed demand per cell and bucket, which exposes systematic over- or under-prediction and the demand range at which the fit degrades, read as spread away from the diagonal (Figure 10).
- **Spatial error.** Error aggregated per hexagon and mapped three ways: absolute error (Figure 11), error relative to each cell's own mean demand (Figure 12), and the share of each cell's day-to-day variation the model explains (Figure 13). The absolute and relative views answer different questions, since the largest absolute errors concentrate downtown purely because demand is largest there, while the relative view is what makes the quiet outer cells legible at all.
- **Head-to-head by location.** The per-hexagon difference $MAE_{NN} - MAE_{SVM}$ mapped directly, so that which model wins where is read off a single diverging scale rather than inferred by comparing two separate colour ramps (Figure 14).
- **Temporal error.** Error by day of week (Figure 15,
 - 1) and, on the hourly grid only, by hour of day (Figure 17, Figure 18), each in absolute terms and relative to mean demand. This separates error that simply tracks demand volume from error concentrated at particular times, such as the overnight hours where demand is near zero and small absolute misses dominate the relative figure. City-wide predicted and actual demand across the full test period is plotted alongside as a sanity check on aggregate level and drift (Figure 19).
- **Error by segment.** Test MAE, relative error and skill score broken down by demand band (zero, 1-10, 11-100, and above 100 trips) and by weather (dry against rainy days), as a table (Table 7) and as relative error per segment (Figure 20). This is the decisive view for fleet planning, because a model that wins on average can still lose on exactly the busiest cells, or on the rainy days, that matter operationally.

4.2.5 Shortfalls and Improvement Levers

Neither family clears the historical mean by a wide margin, and the reasons split into three groups: limits specific to the SVM, limits specific to the network, and limits the two share.

For the **support vector machine**, one failure mode is specific and fixable. The residual model is worse than the baseline precisely on cells whose demand turns out to be zero, because the correction term does not shrink to zero where it should, and zero rows dominate most levels. A floor on the reconstructed correction, or a gate that decides zero against non-zero before

regressing magnitude, would both be worth testing. Smaller gains could come from rebalancing the sample (a natural-prevalence draw performed worse) or from a larger one, though SVR’s quadratic fit cost is what forced a balanced subsample in the first place, so more rows trade directly against the breadth of the tuning search.

The **network’s** limits are ones of search breadth rather than a diagnosed failure. A single feature set was used at every level, so unlike the SVM there is no evidence on whether weather or POI features help or hurt the network at any given level; repeating the four-way feature-set comparison is the most direct improvement lever. The architecture search was bounded the same way, covering layer width, learning rate and dropout across 24 configurations while holding batch size, activation function and epoch budget fixed. No permutation-importance or comparable interpretability check was run for the network, only for the direct SVR, so whether the weather and POI blocks contribute at all is currently unknown.

Both families fail on the same grids. Census/hourly and census/daily, the sparsest and most zero-inflated levels, defeat each of them, which points at the data rather than at either model choice. More generally, **base_demand** already carries most of the signal that a scenario-conditioned framing makes available, so both models compete for a thin margin on top of an already strong baseline. The levers that would widen that margin are therefore shared: weather interacted with hour and location rather than entered additively, weather resolved spatially instead of drawn from a single citywide coordinate (Section 3), and POI expressed as distances to the nearest attractor rather than as counts inside the cell.

4.3 Smart Charging (Reinforcement Learning)

We frame the taxi driver’s home charging as a Markov Decision Process and solve it with Q-learning. The agent sets the charging power in 15-minute steps between the 2 p.m. arrival and 4 p.m. departure, trading off recharging cost against the risk of not meeting the (stochastic) energy demand for the working day.

4.3.1 Problem Formulation (MDP)

4.3.2 Environment (Discrete-Event Simulation)

4.3.3 Q-Learning Agent

4.3.4 Training

4.3.5 Learned Policy

The heatmap below shows the learned charging policy: for each battery level and timestep, the charging action the agent has learned to take.

4.3.6 Evaluation

5 Discussion and Outlook

5.1 Summary of Findings

5.2 Implications for the Client

5.3 Limitations

5.4 Future Work

A Additional Figures and Tables

This appendix collects supporting figures and tables that are not reproduced in the main body of the report, organized by the notebook that produces them (see the pipeline overview in the project `CLAUDE.md`).

A.1 Taxi Data Preparation (1.1)

No additional labeled figures or tables from this notebook.

A.2 Weather Data Preparation (1.2)

No additional labeled figures or tables from this notebook.

A.3 Points of Interest Data Preparation (1.3)

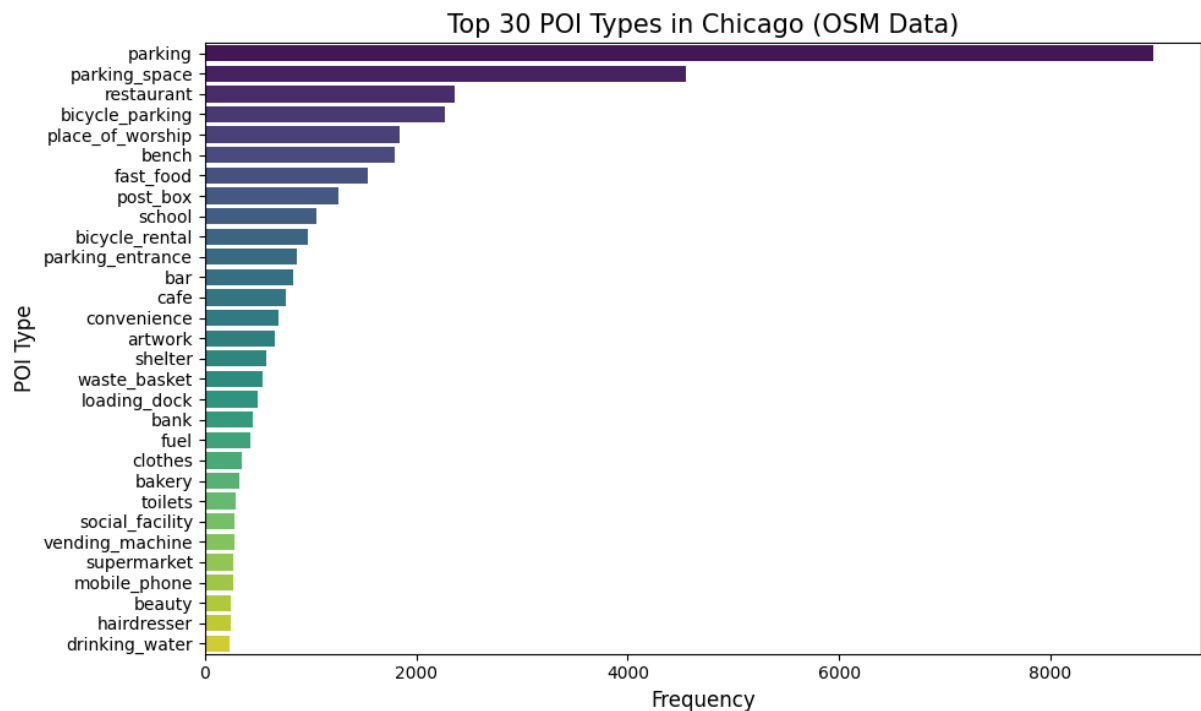


Figure 1: Top 30 most frequent raw OSM POI types in Chicago, before category mapping.

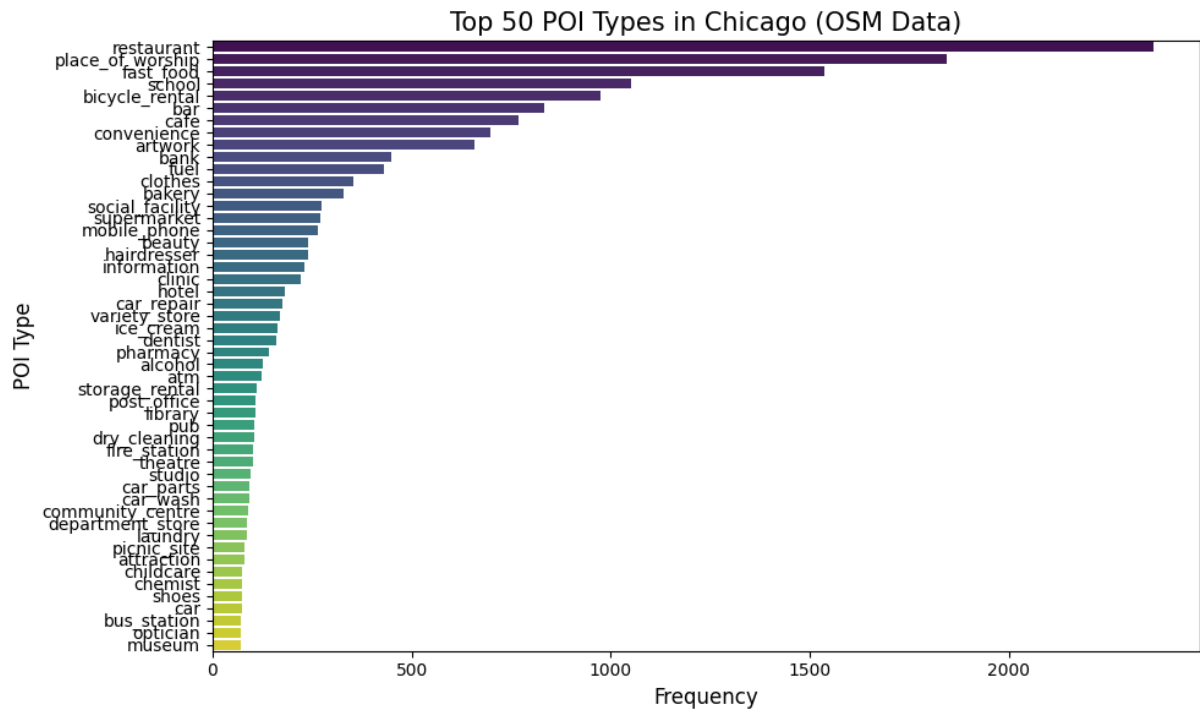


Figure 2: Top 50 most frequent POI types remaining after category filtering (unmapped ‘other’ types dropped).

A.4 Temporal-Spatial Discretization (1.4)

No additional labeled figures or tables from this notebook.

A.5 Descriptive Analysis: Spatial (2.1)

No additional labeled figures or tables from this notebook.

A.6 Descriptive Analysis: Temporal (2.2)

No additional labeled figures or tables from this notebook.

A.7 Descriptive Analytics: Weather (2.4)

No additional labeled figures or tables from this notebook.

A.8 SVR Hyperparameter Search Grid (3.2)

Both SVR notebooks share the identical two-stage search defined in 3.2 `PredictionCommonGround.ipynb` so the direct and residual results stay directly comparable. The grid search runs only for the kernel that wins the scan, not for all three.

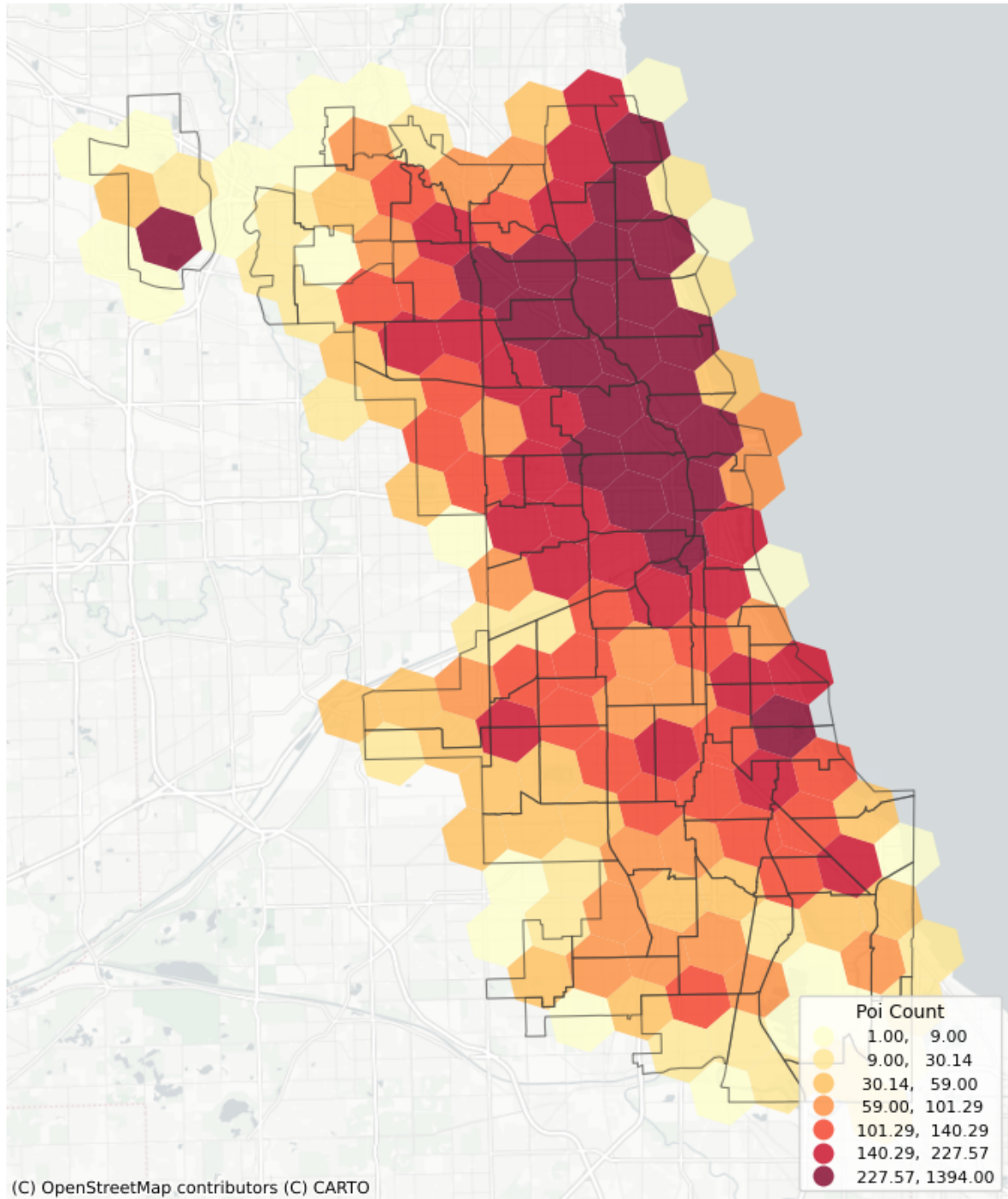
Chicago — POI Count by H3 Hexagon (resolution 7)

Figure 3: Total POI count per H3 hexagon (resolution 7).

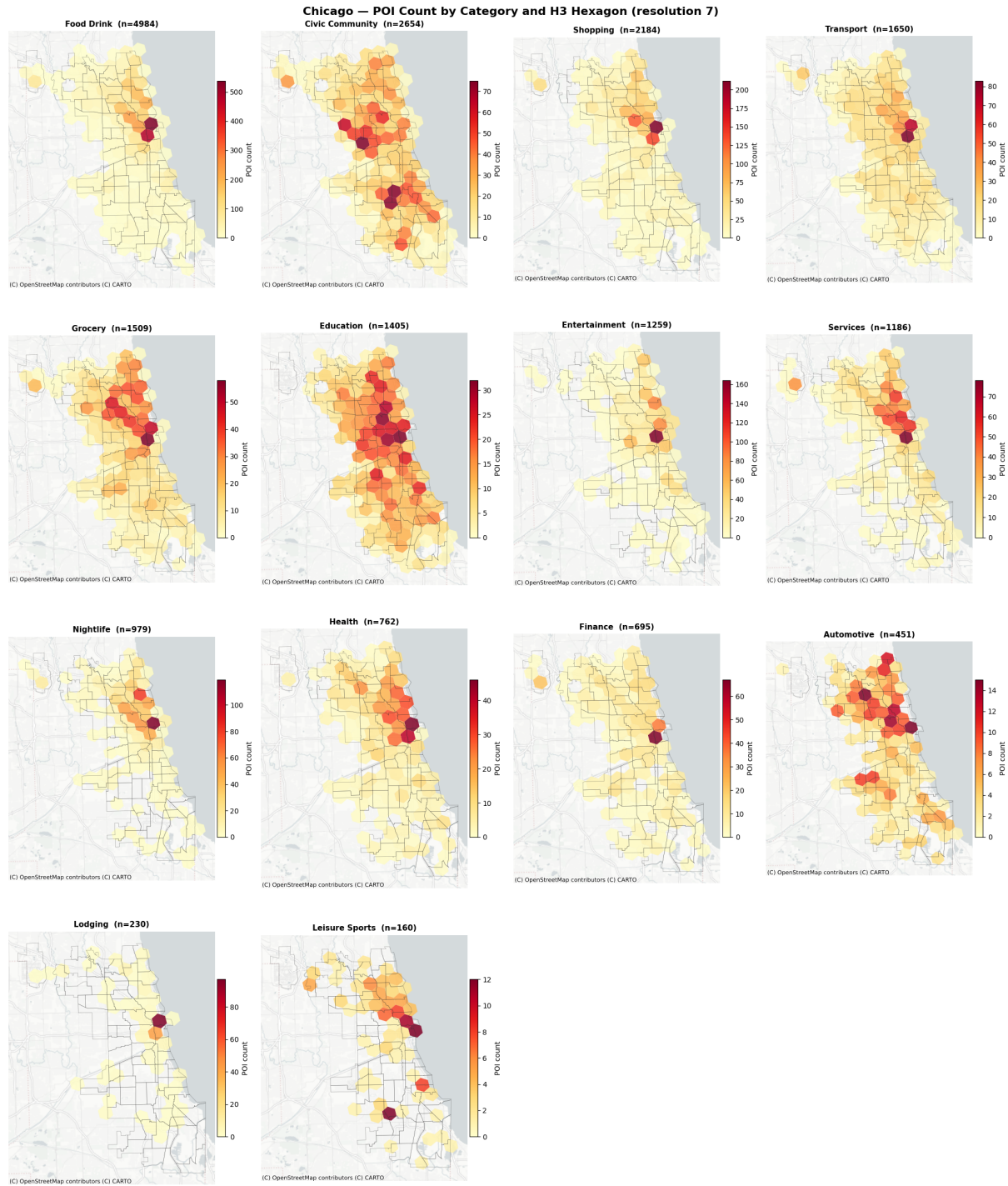


Figure 4: POI count per H3 hexagon (resolution 7), faceted by category.

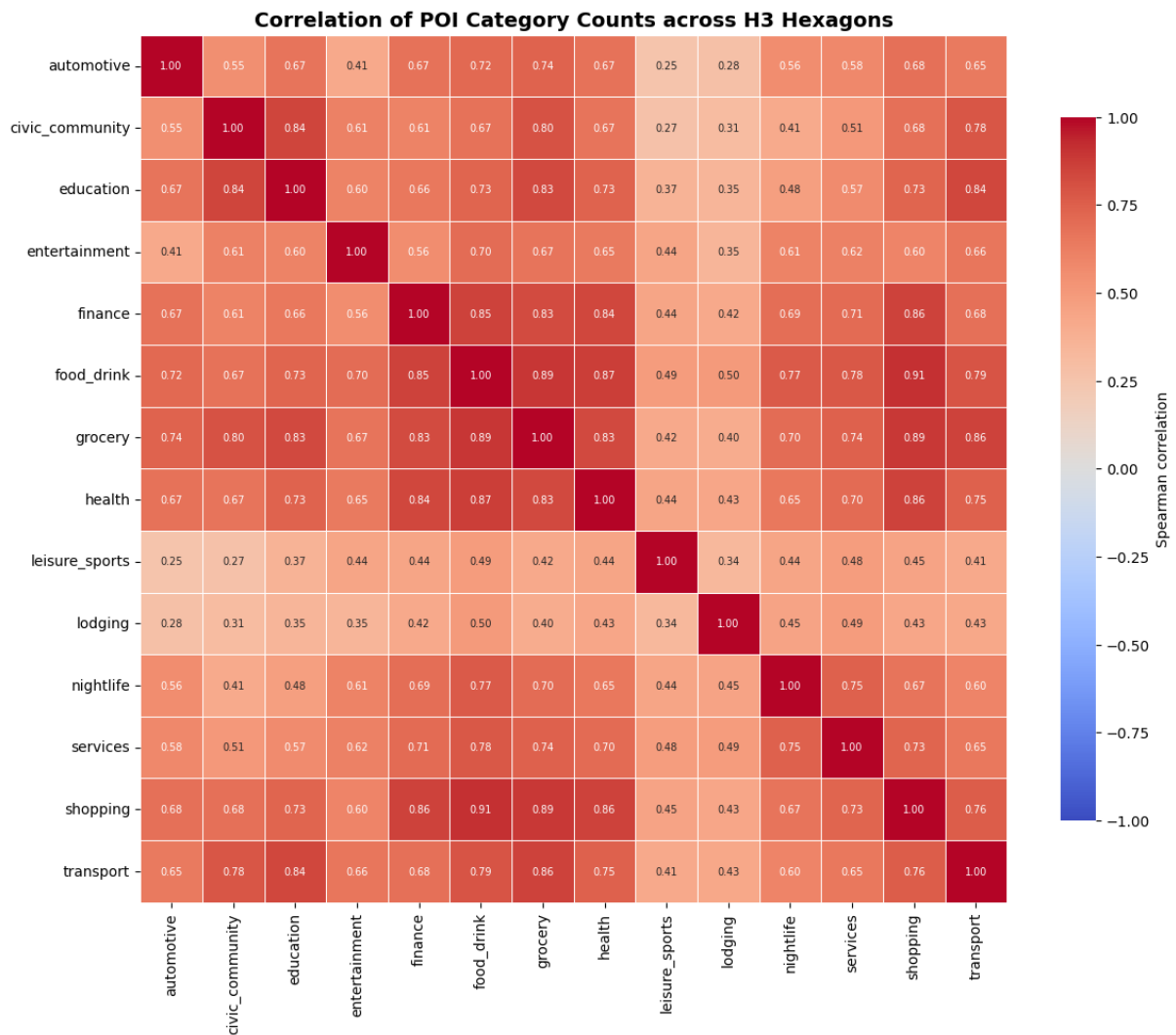


Figure 5: Spearman correlation of POI category counts across H3 hexagons (resolution 7).

Table 1: Two-stage SVR hyperparameter search grid, shared by the direct (3.2a) and residual (3.2b) models.

Stage	Setting	Values searched
Kernel scan	Kernels	<code>linear</code> , <code>poly</code> , <code>rbf</code>
	Held fixed during the scan	<code>C = 1</code> , <code>gamma = "scale"</code>
Grid search	Selection criterion	Validation R^2 in count space
	Training sample	4,000 rows per class (8,000 total)
	<code>C</code>	1, 10, 50
	<code>gamma</code>	"scale", 0.05, 0.2
	<code>degree</code> (polynomial kernel only)	2, 3
	Cross-validation	3-fold
Feature sets	Selection criterion	Count-space MAE
	Training sample	8,000 rows per class (16,000 total)
	Candidates	<code>basic</code> , <code>basic+poi</code> , <code>basic+weather</code> , <code>basic+poi+weather</code>
	Selection criterion	Validation R^2 in count space
Scoring	Validation subsample	50,000 rows; the level's winner is re-scored on the full validation split

The four candidate feature sets are built from the same blocks at every level. `basic` is hour (hourly grids only) plus calendar, location and history; the other three add the POI and weather blocks to it.

Table 2: Feature blocks composing the four candidate sets. Resulting widths are 10 / 24 / 15 / 29 features on daily grids and 12 / 26 / 17 / 31 on hourly grids, for `basic` / `basic+poi` / `basic+weather` / `basic+poi+weather` respectively.

Block	Features	Count
Hour (hourly grids only)	<code>hour_sin</code> , <code>hour_cos</code>	2
Calendar	<code>month_sin</code> , <code>month_cos</code> , <code>is_weekend</code> , <code>is_holiday</code> , <code>is_near_holiday</code> , <code>day_of_week</code>	6
Location	<code>lat</code> , <code>lon</code> , <code>distance_to_loop</code>	3
History	<code>base_demand</code>	1
POI	14 <code>poi_cat_*</code> category counts	14
Weather	<code>2m_temp_c</code> , <code>total_precip_mm</code> , <code>snow_cov</code> , <code>snow_depth</code> , <code>wind_speed</code>	5

A.9 SVR: Direct Model (3.2a)

Table 3: Winning SVR per level, with skill against the historical-mean base

	spatial	temporal	feature_set	kernel	C	gamma	val_MAE	val_RMSE	val_R2
0	h3_7	daily	basic+poi	rbf	50	scale	15.7997	88.8310	0.9494
1	community	daily	basic+poi	rbf	10	scale	37.4429	139.8964	0.9578

Table 3: Winning SVR per level, with skill against the historical-mean base

	spatial	temporal	feature_set	kernel	C	gamma	val_MAE	val_RMSE	val_R2
2	census	daily	basic+poi	rbf	10	scale	4.0758	27.3988	0.9341
3	community	hourly	basic+poi	rbf	1	scale	2.7382	10.3958	0.9133
4	h3_7	hourly	basic	rbf	10	0.05	0.9858	6.2626	0.9148
5	census	hourly	basic+poi+weather	rbf	1	scale	0.3302	2.0741	0.8795

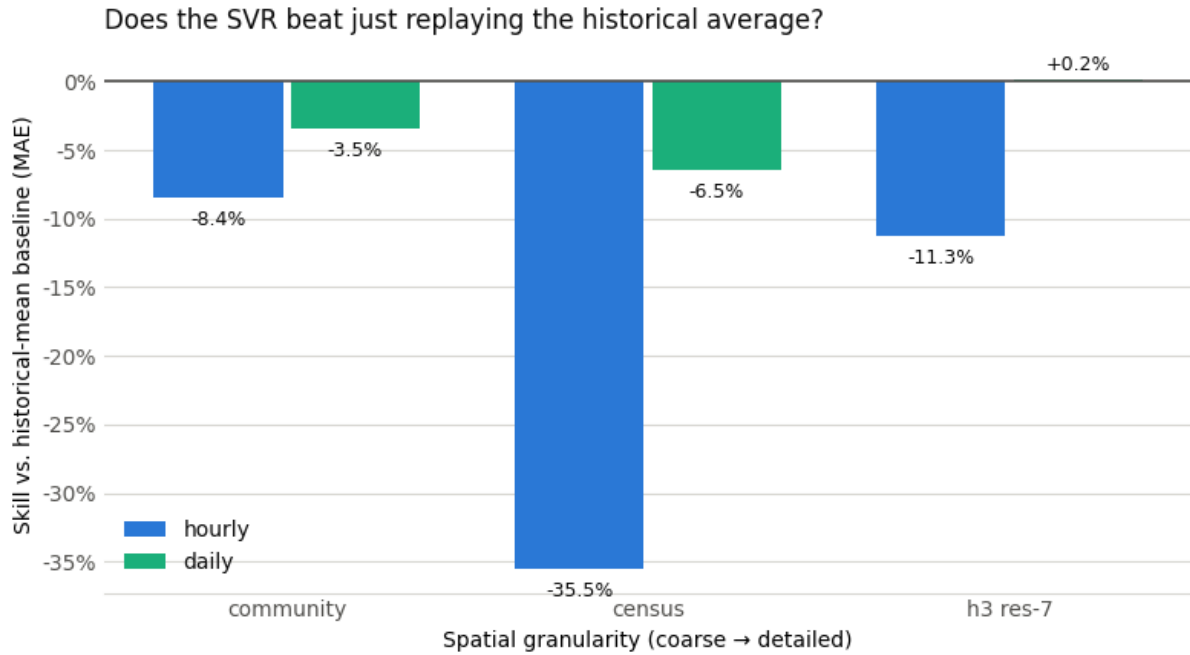


Figure 6: SVR skill against the historical-mean baseline, by spatial and temporal granularity.

A.10 SVR: Residual-on-Baseline Model (3.2b)

Table 4: Winning residual-on-baseline SVR per level, with skill against the historical-mean baseline.

	spatial	temporal	feature_set	kernel	C	gamma	val_MAE	val_RMSE	val_R2
0	community	daily	basic+weather	linear	1	scale	34.2541	118.3257	0.9698
1	h3_7	daily	basic+poi	linear	1	scale	15.6367	84.9372	0.9538
2	census	daily	basic+poi+weather	rbf	1	scale	3.8174	26.1691	0.9399
3	h3_7	hourly	basic+weather	linear	1	scale	0.8838	5.5285	0.9336
4	community	hourly	basic+poi+weather	rbf	1	scale	2.5498	9.2293	0.9316
5	census	hourly	basic	linear	1	scale	0.3221	1.9674	0.8915

Feature importance: reliance vs. necessity (= block not in this model's feature set)

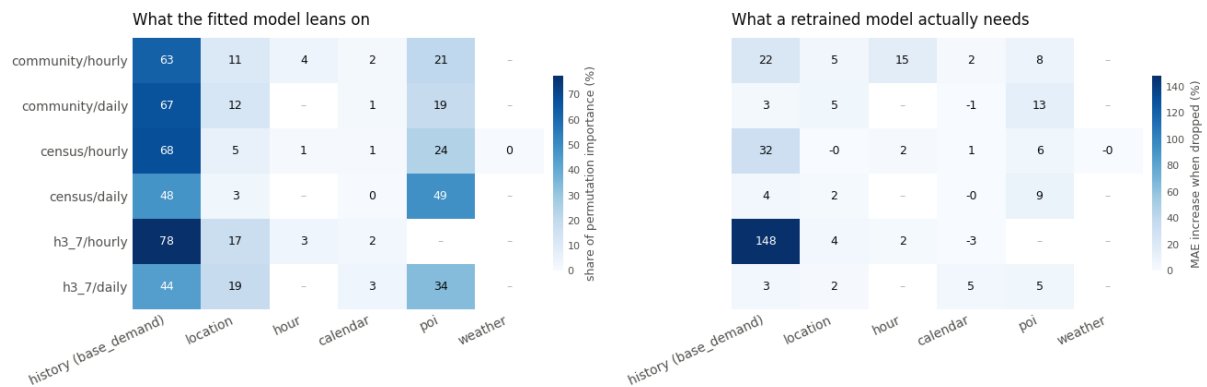


Figure 7: Left: what the fitted model leans on (grouped permutation importance, as each block's share of the level's total). Right: what a retrained model actually needs (drop-column ablation, % increase in MAE when the block is removed). The two disagree sharply on base_demand — the fitted model depends on it almost entirely, yet a model refitted without it loses very little, because lat/lon/hour/day-of-week can reconstruct it.

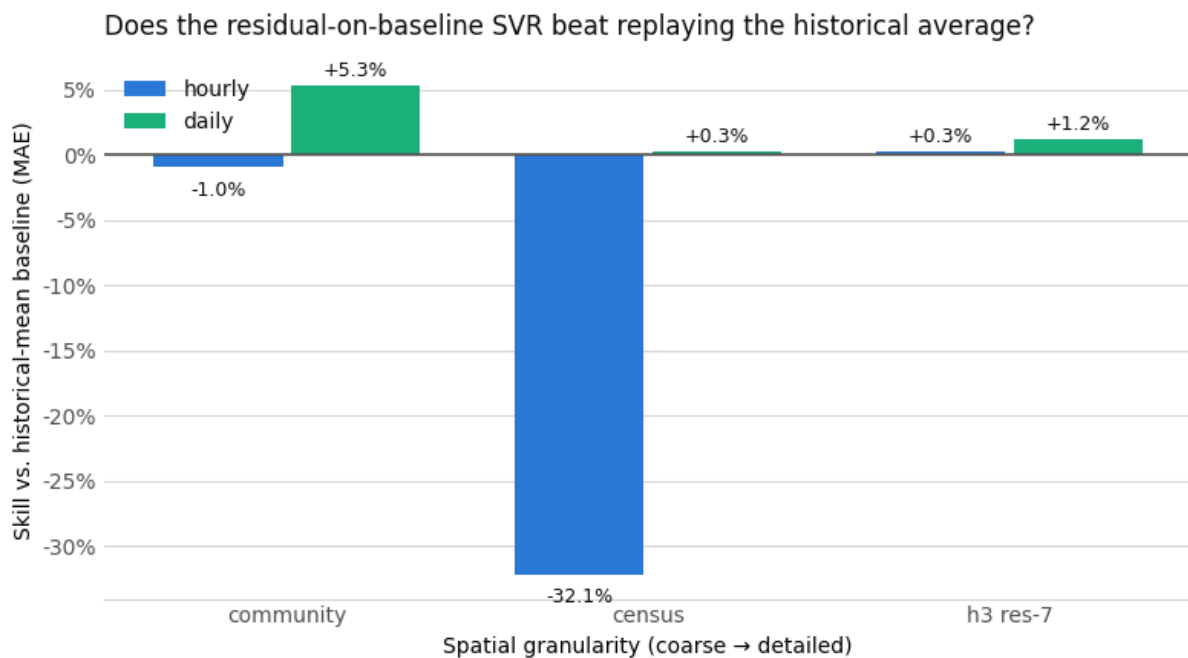


Figure 8: Residual-on-baseline SVR skill against the historical-mean baseline, by spatial and temporal granularity.

Table 5: Direct SVR (3.2a) vs. residual-on-baseline SVR trained from scratch (this notebook), per level, on the validation split.

	spatial	temporal	feature_set	kernel	skill_MAE_direct	skill_MAE_residual	delta_
0	h3_7	hourly	basic+weather	linear	-0.1125	0.0026	0.1151
1	community	daily	basic+weather	linear	-0.0347	0.0534	0.0881
2	community	hourly	basic+poi+weather	rbf	-0.0844	-0.0098	0.0746
3	census	daily	basic+poi+weather	rbf	-0.0646	0.0029	0.0675
4	census	hourly	basic	linear	-0.3547	-0.3214	0.0333
5	h3_7	daily	basic+poi	linear	0.0019	0.0122	0.0103

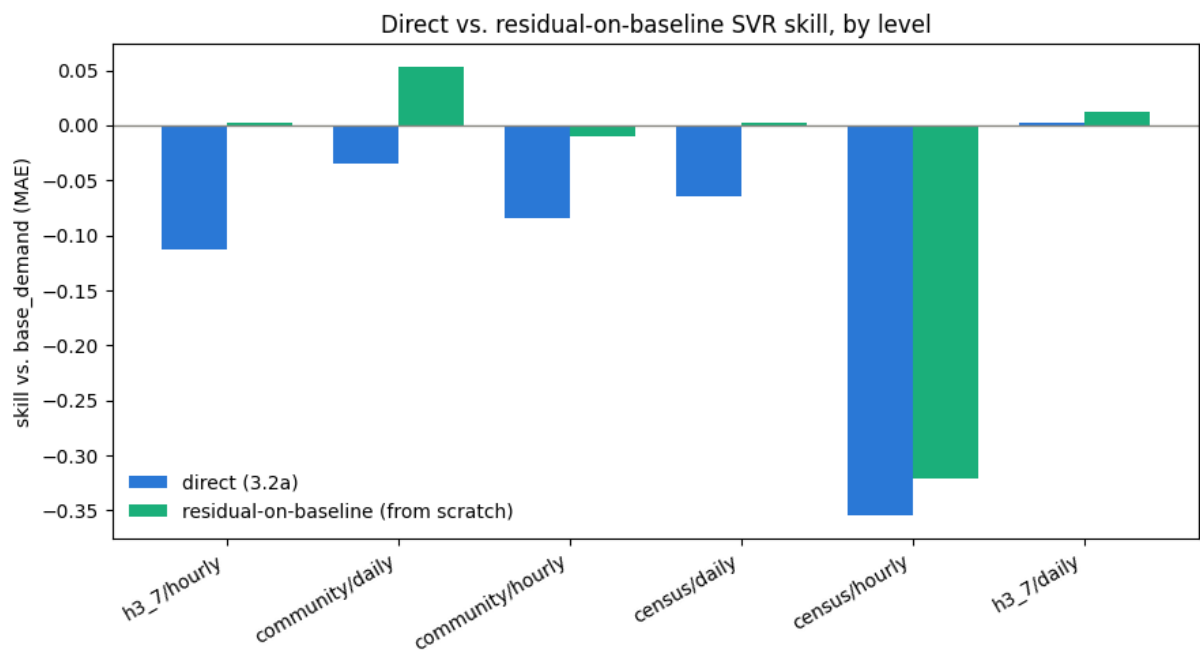


Figure 9: Direct (3.2a) vs. residual-on-baseline (this notebook) SVR skill, both independently tuned, per level.

A.11 Deep Learning Prediction (3.3)

No additional labeled figures or tables from this notebook.

A.12 SVM vs. Neural Network Performance Comparison (3.4)

H3_7/daily

H3_7/hourly

--- h3_7/daily ---

Table 6: Overall TEST performance at h3_7/daily and h3_7/hourly.

(a) h3_7/daily			
	MAE	RMSE	R2
Model			
Baseline	21.3913	121.0296	0.9061
Best SVM (Residual SVR)	20.6391	112.3737	0.9190
Neural Network (FNN)	20.1520	118.8802	0.9094

(b) h3_7/hourly			
	MAE	RMSE	R2
Model			
Baseline	1.0678	7.4296	0.8805
Best SVM (Residual SVR)	1.0165	6.4802	0.9091
Neural Network (FNN)	0.9767	6.4700	0.9094

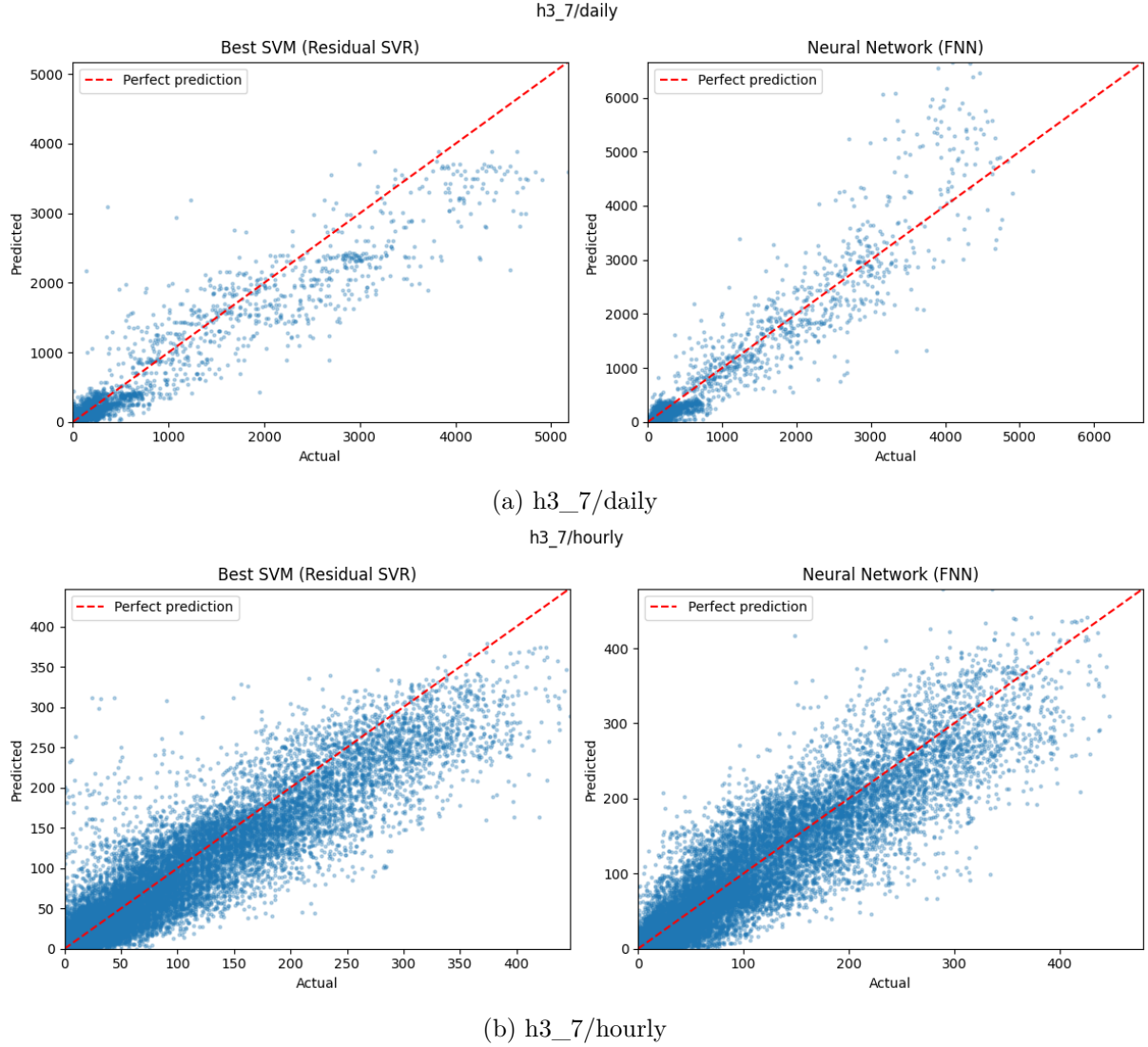
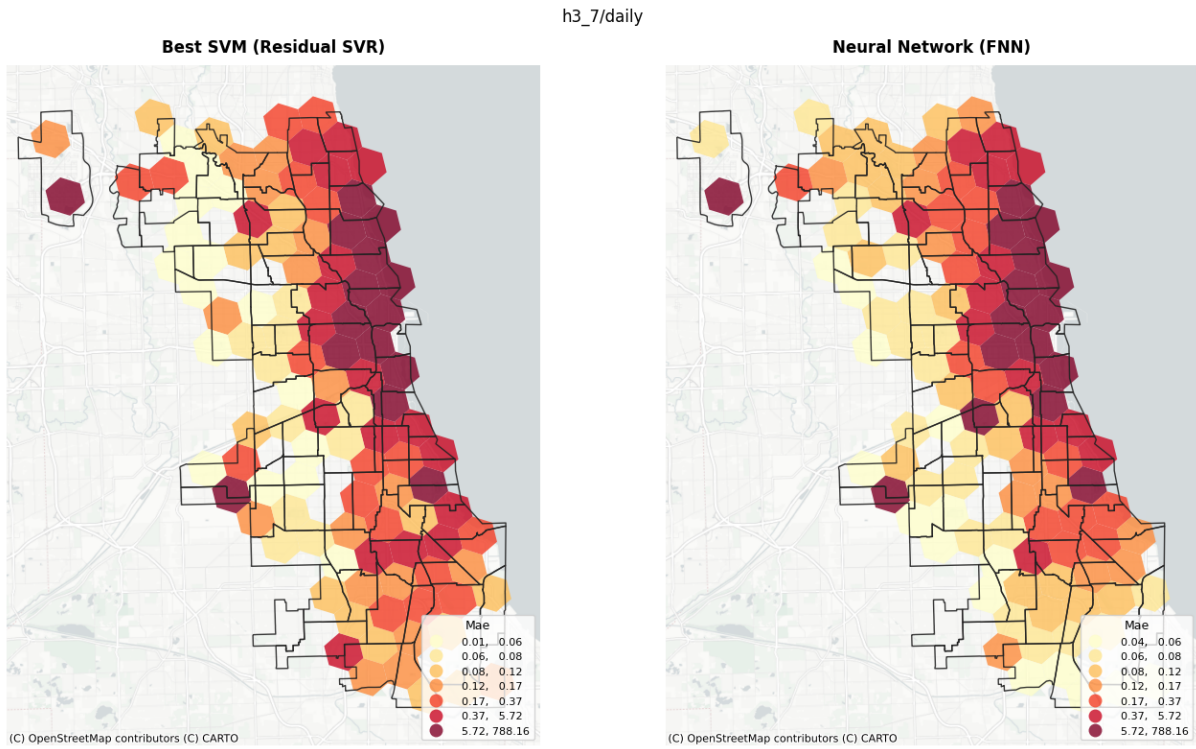
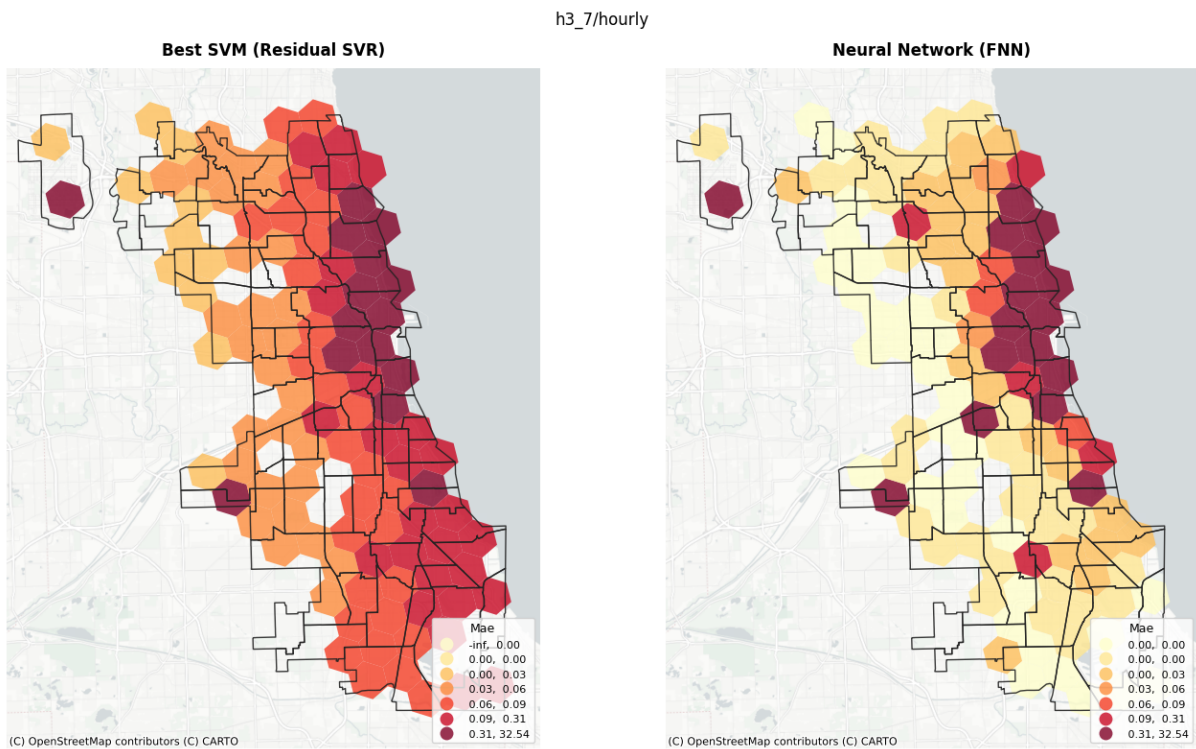


Figure 10: Actual vs. predicted demand per temporal resolution.

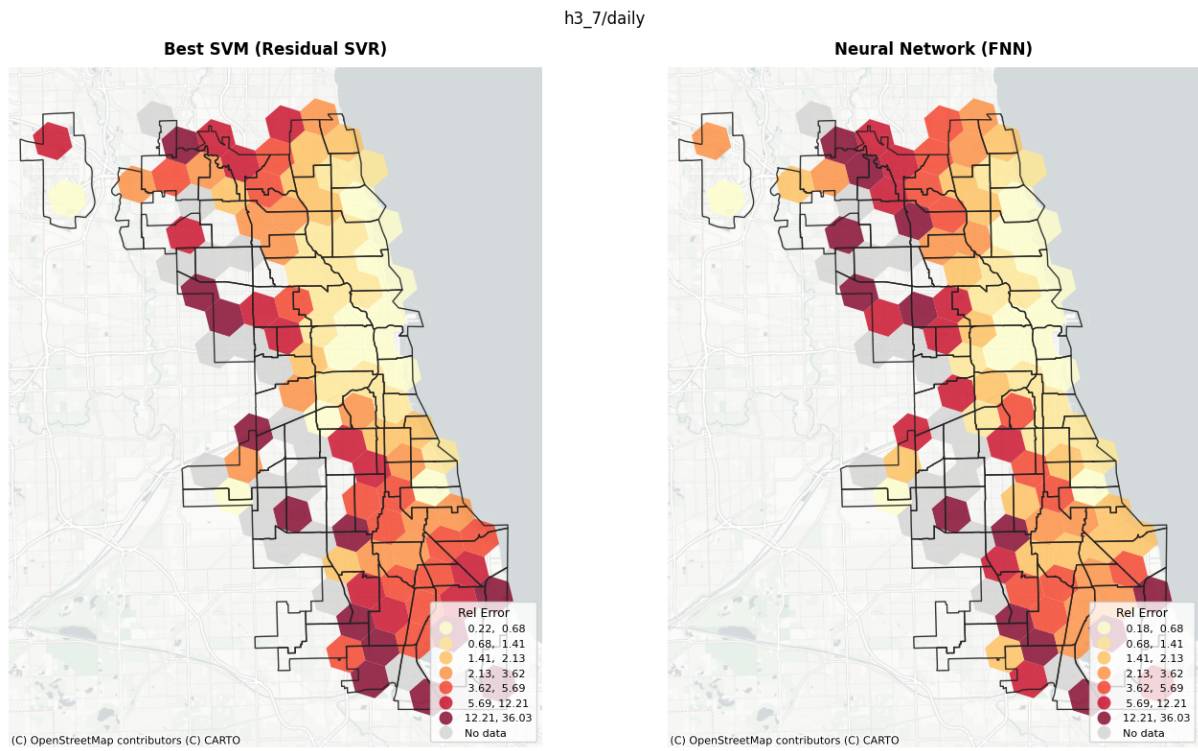


(a) h3_7/daily

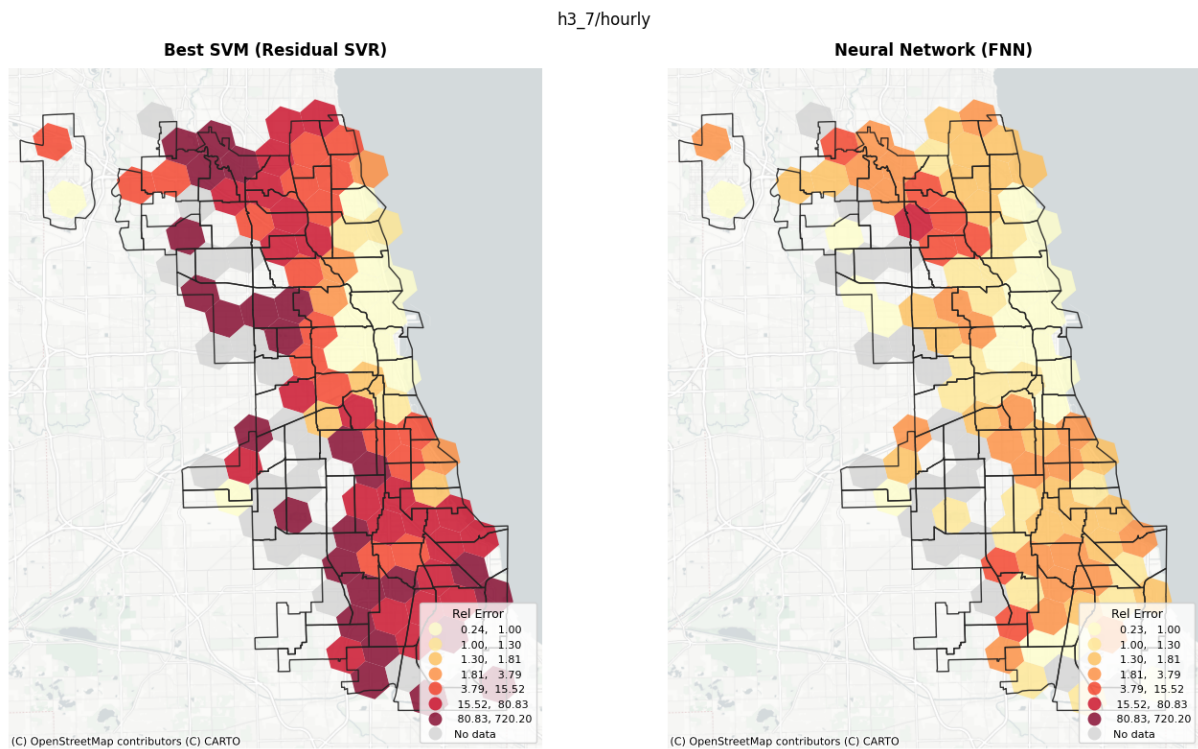


(b) h3_7/hourly

Figure 11: MAE per hexagon per temporal resolution.

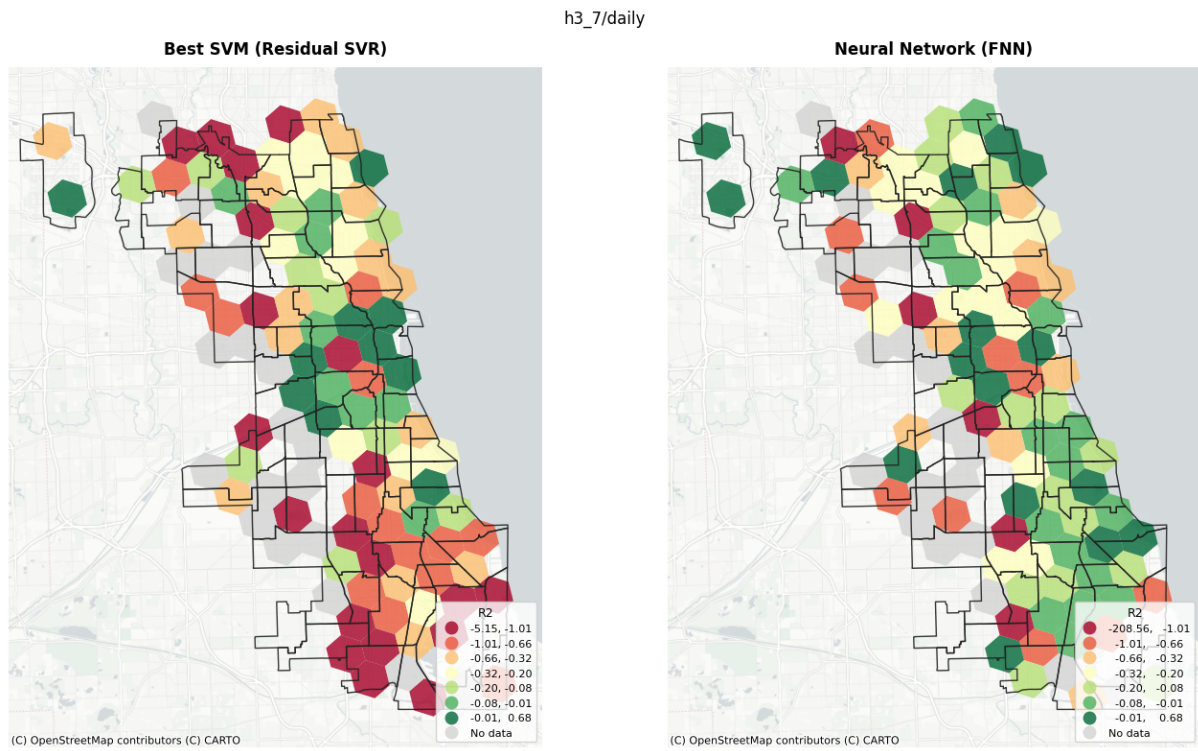


(a) h3_7/daily

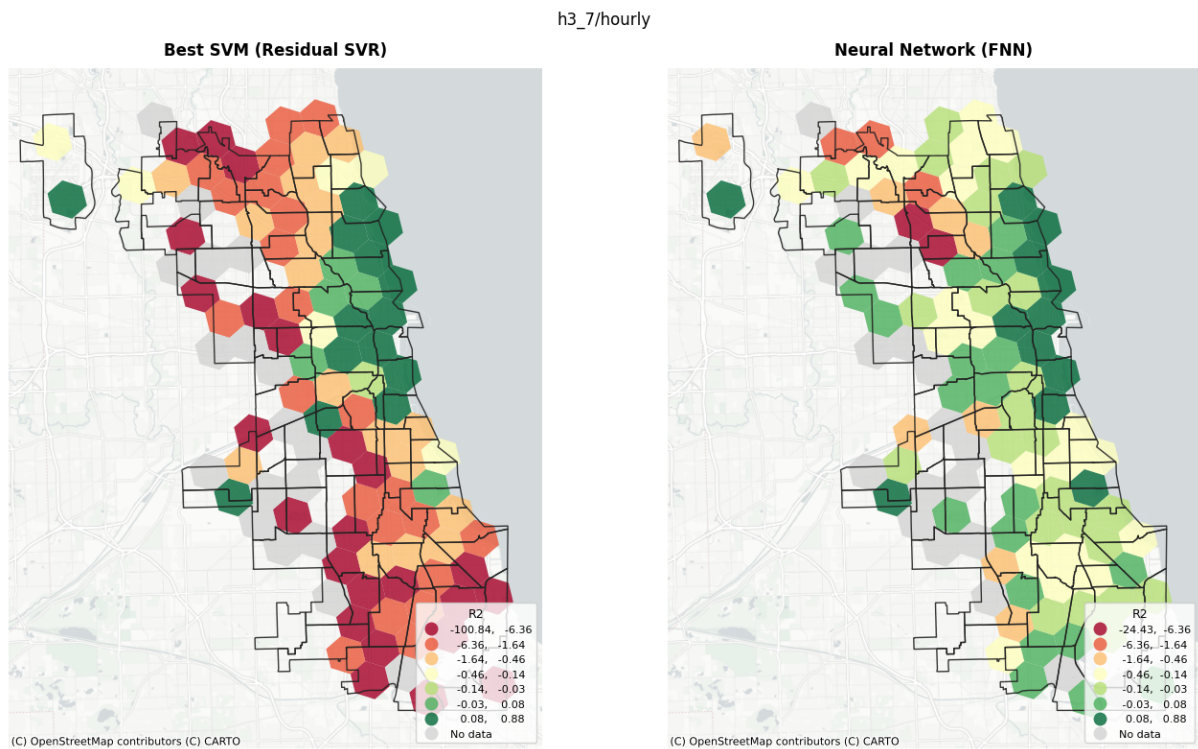


(b) h3_7/hourly

Figure 12: Relative error per hexagon per temporal resolution.

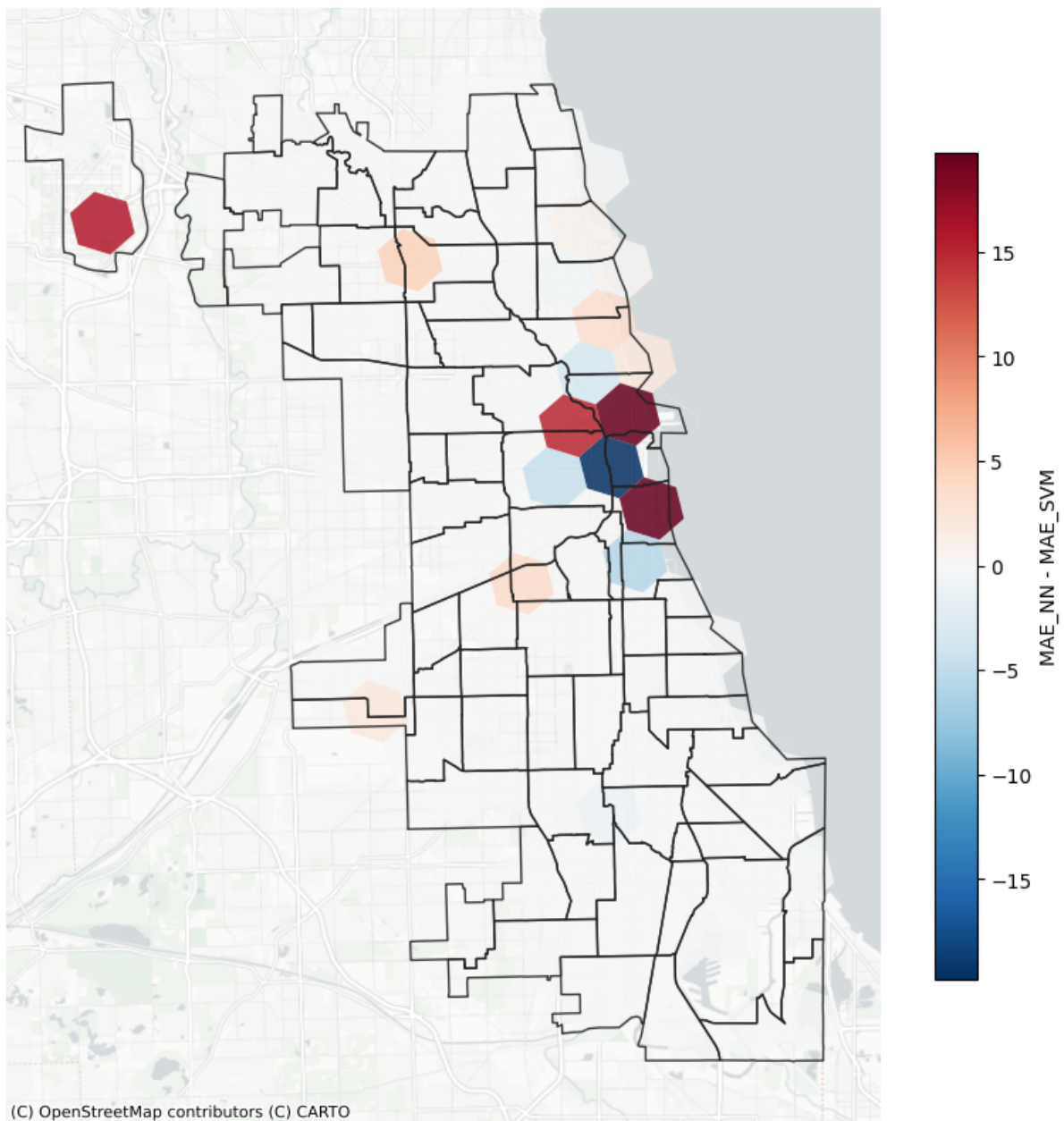


(a) h3_7/daily

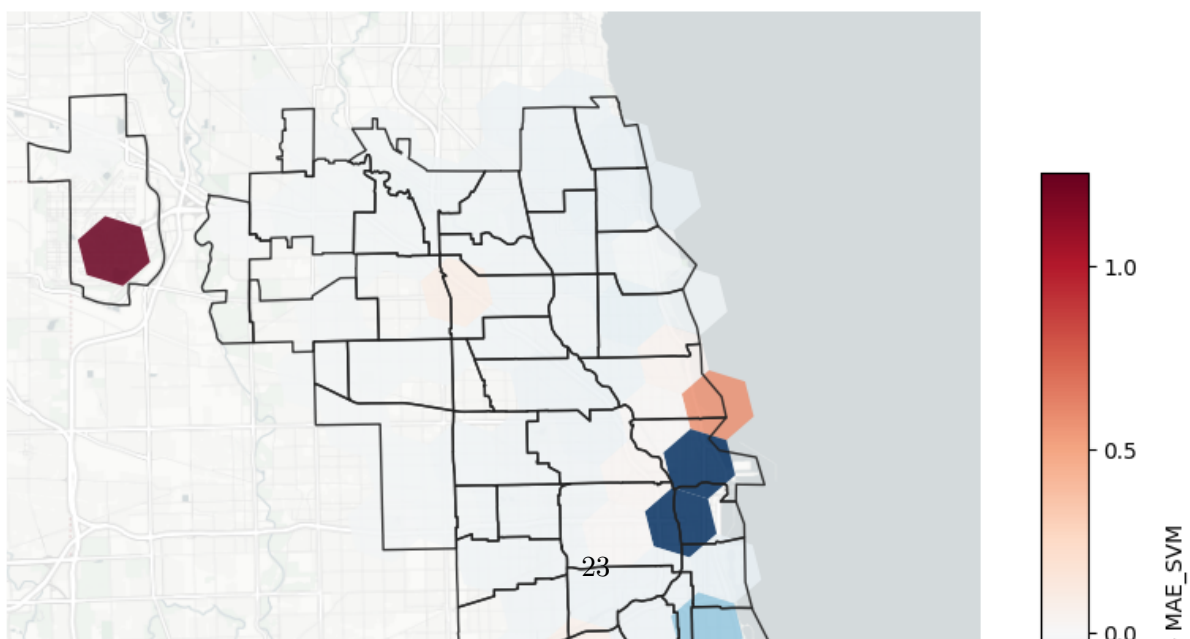


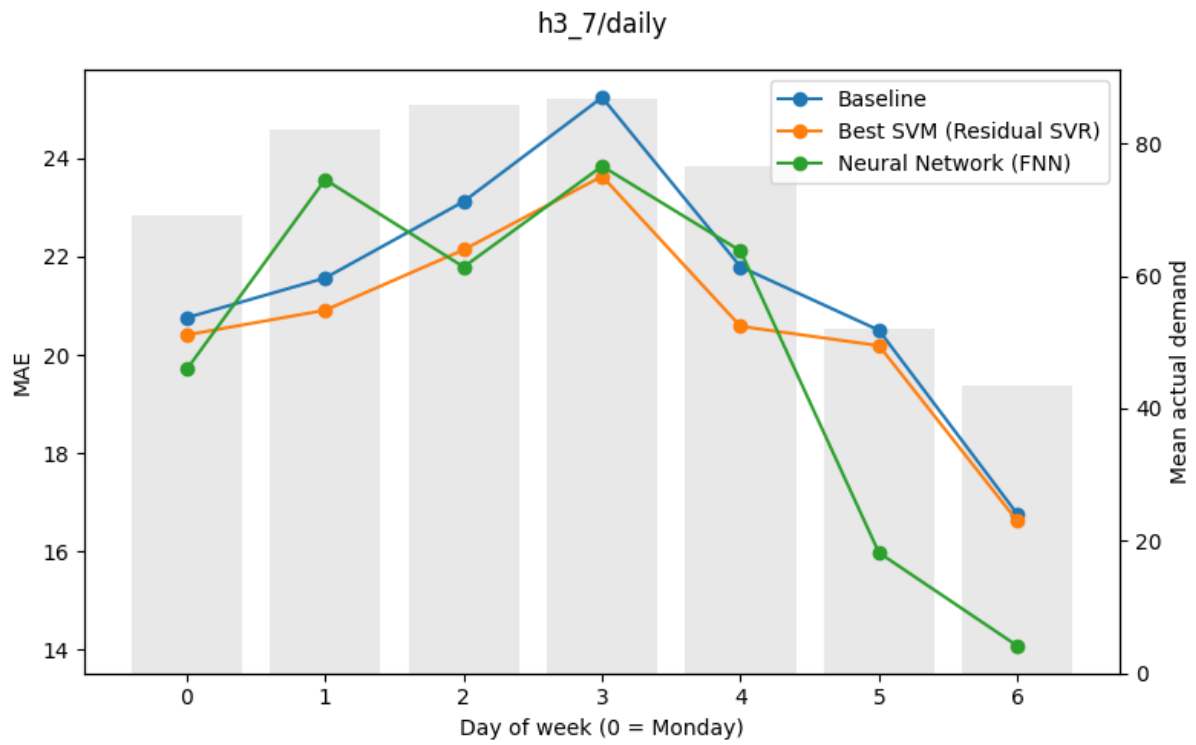
(b) h3_7/hourly

Figure 13: R^2 per hexagon per temporal resolution.

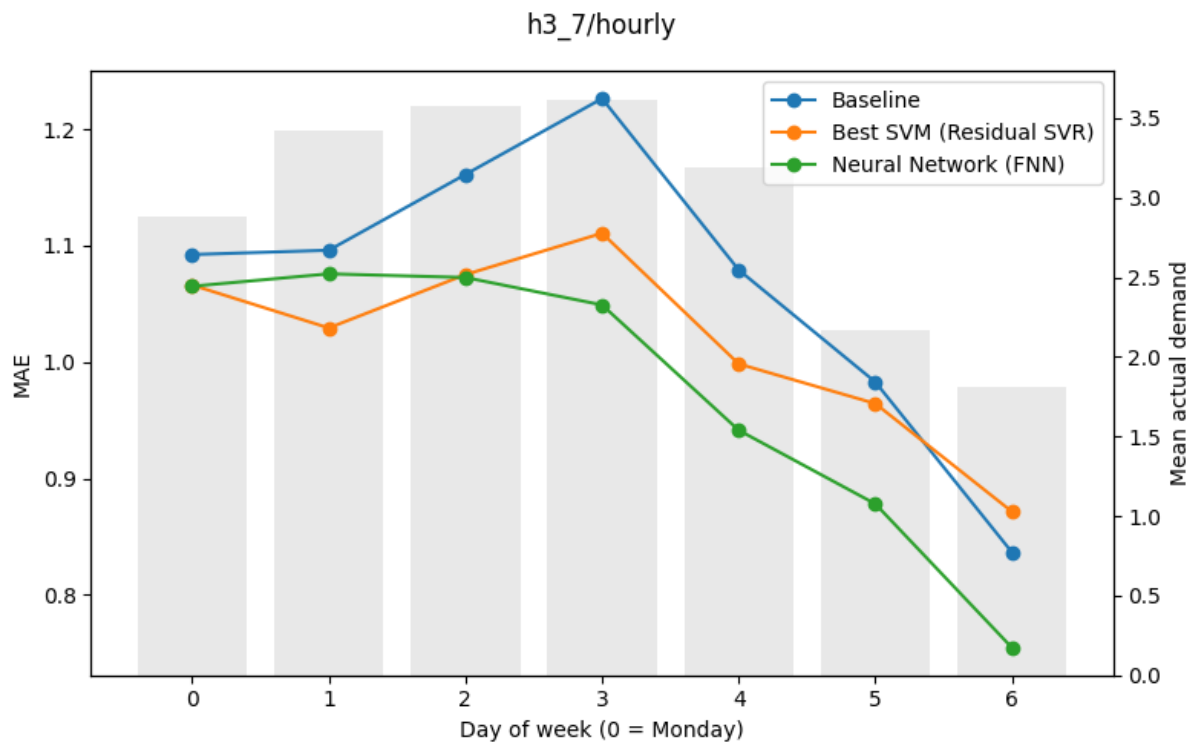
h3_7/daily — Where does SVM beat FNN?

(a) h3_7/daily

h3_7/hourly — Where does SVM beat FNN?

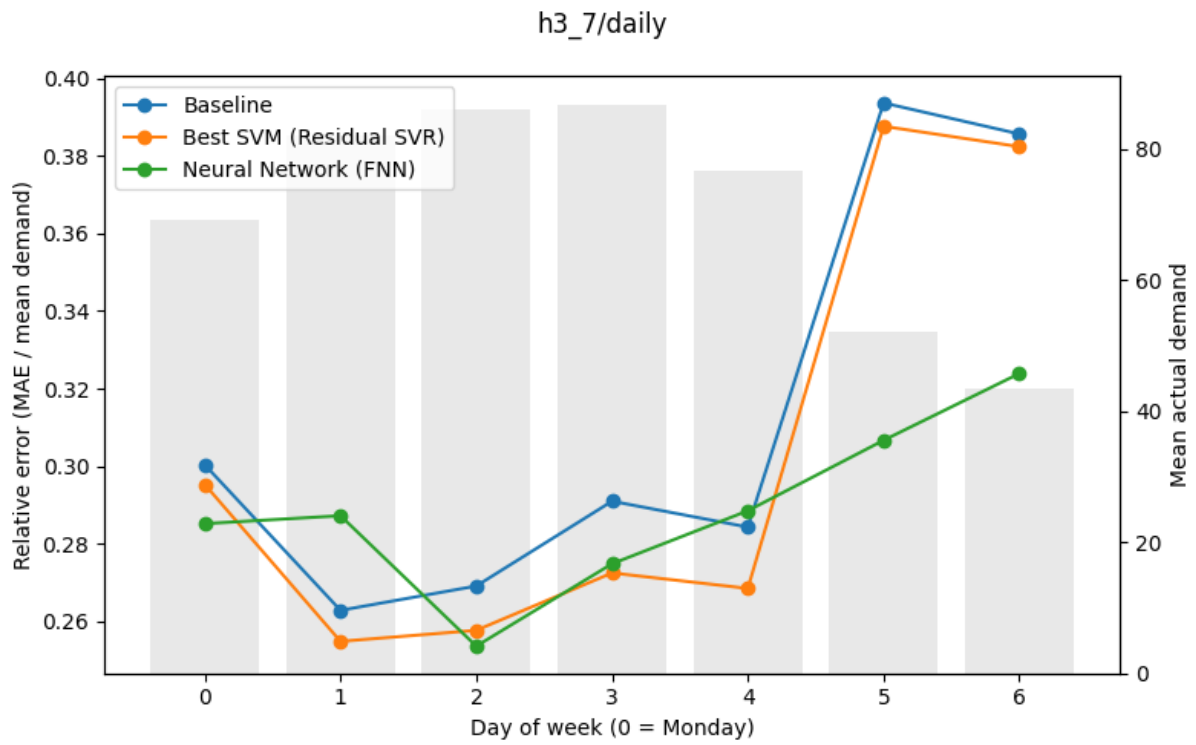


(a) h3_7/daily

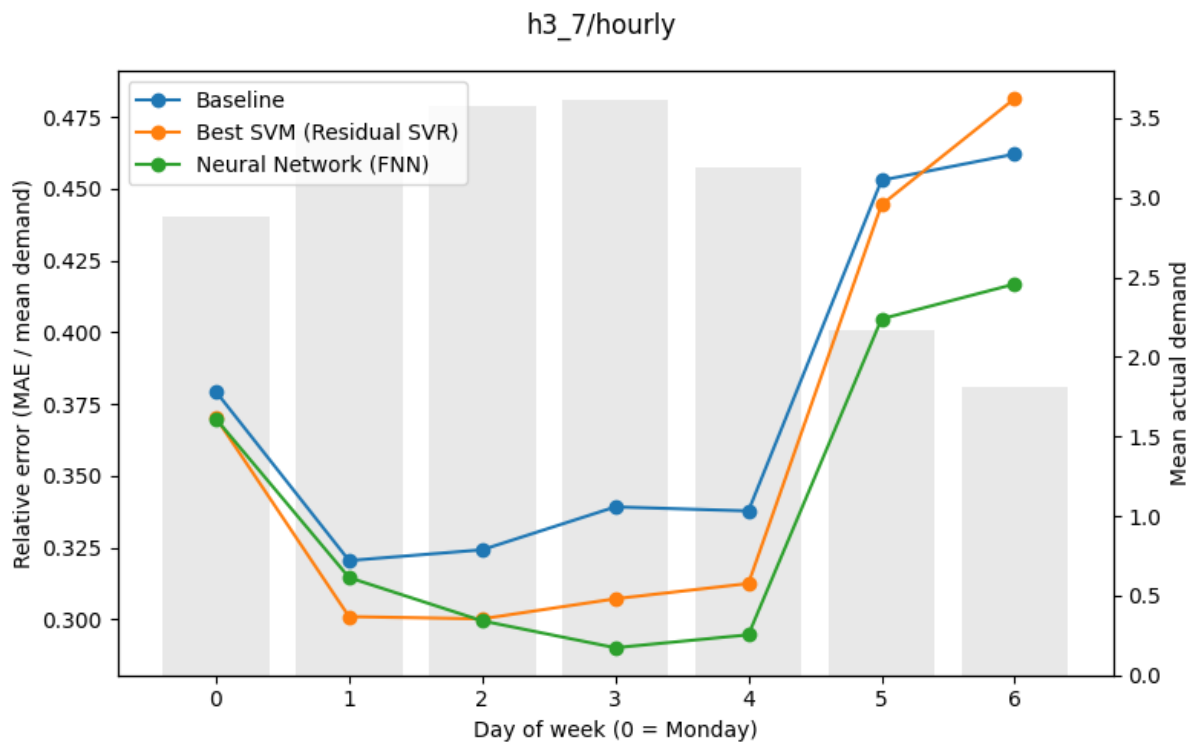


(b) h3_7/hourly

Figure 15: Test MAE by day of week per temporal resolution.



(a) h3_7/daily



(b) h3_7/hourly

Figure 16: Test relative error (MAE / mean demand) by day of week per temporal resolution.

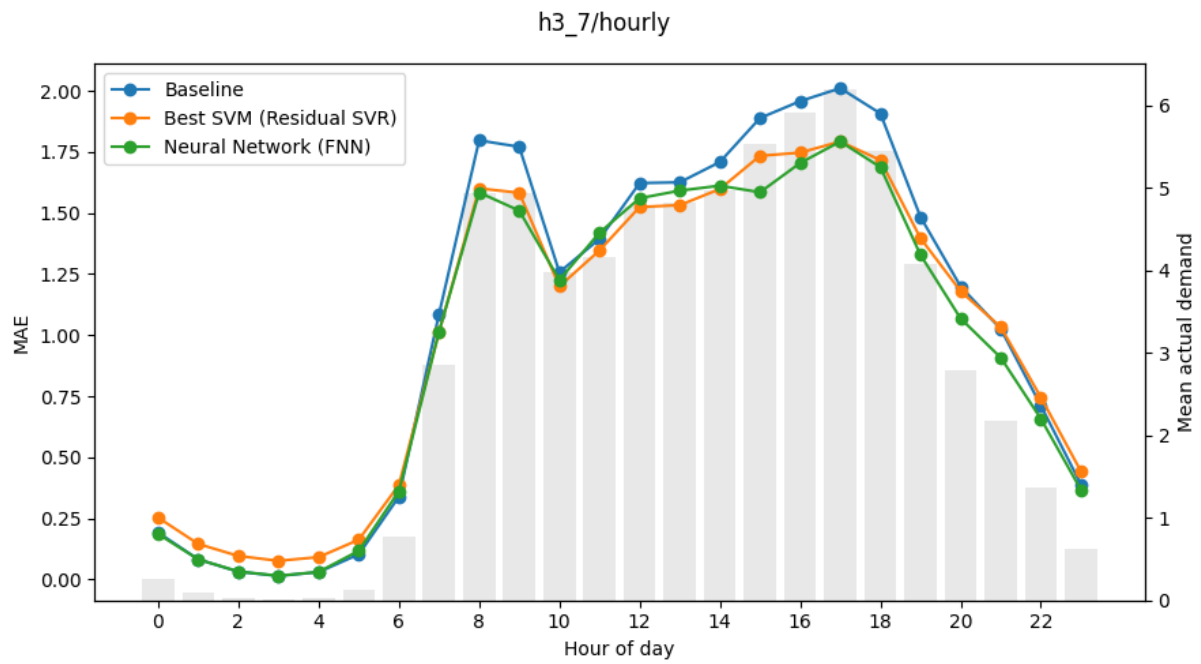


Figure 17: Test MAE by hour of day (h3_7/hourly).

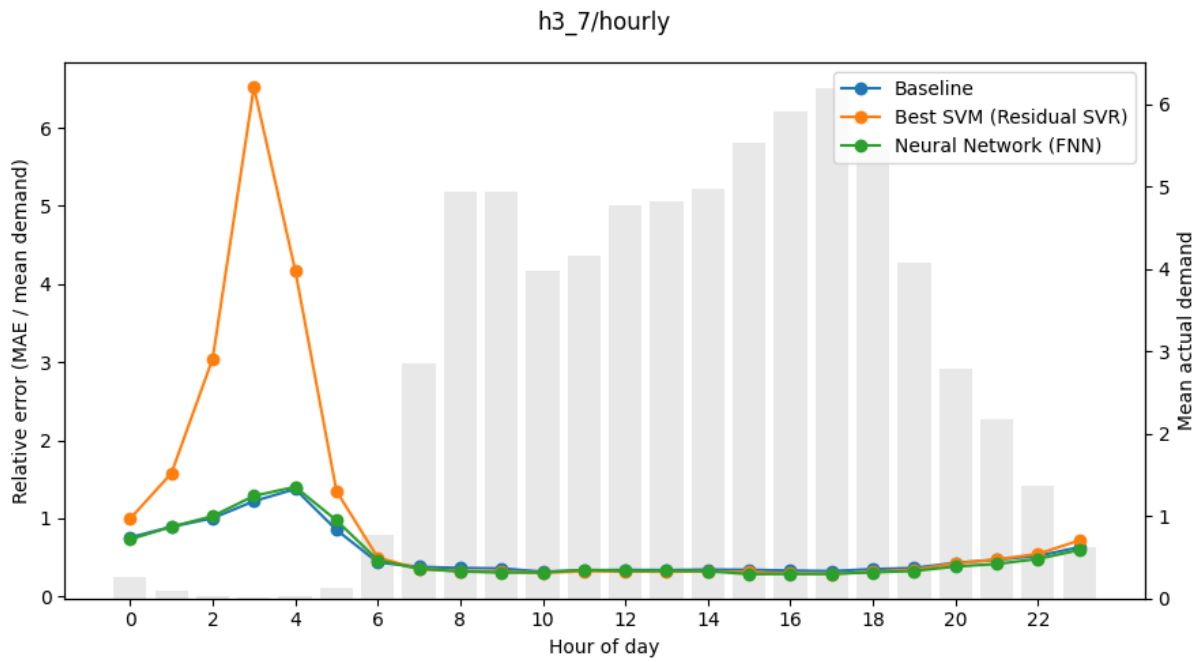
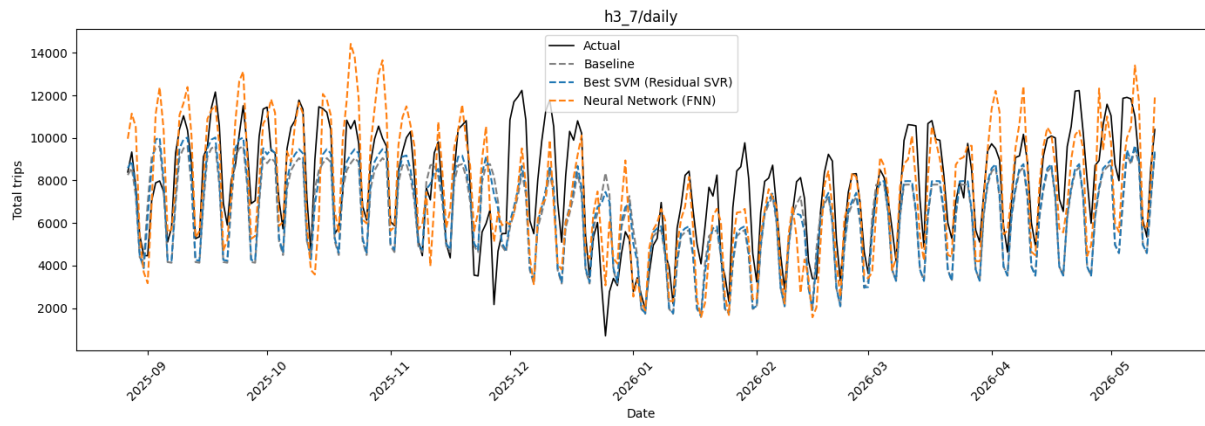
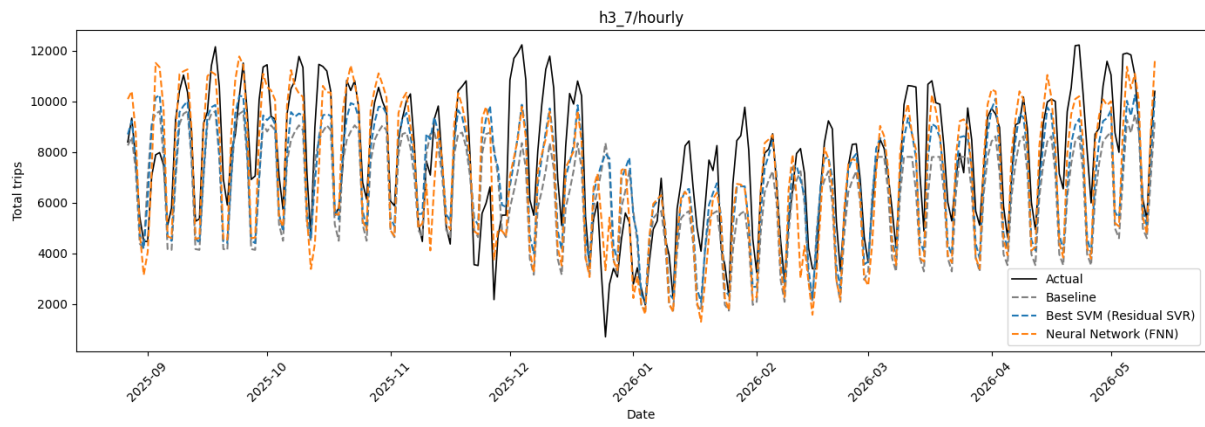


Figure 18: Test relative error (MAE / mean demand) by hour of day (h3_7/hourly).



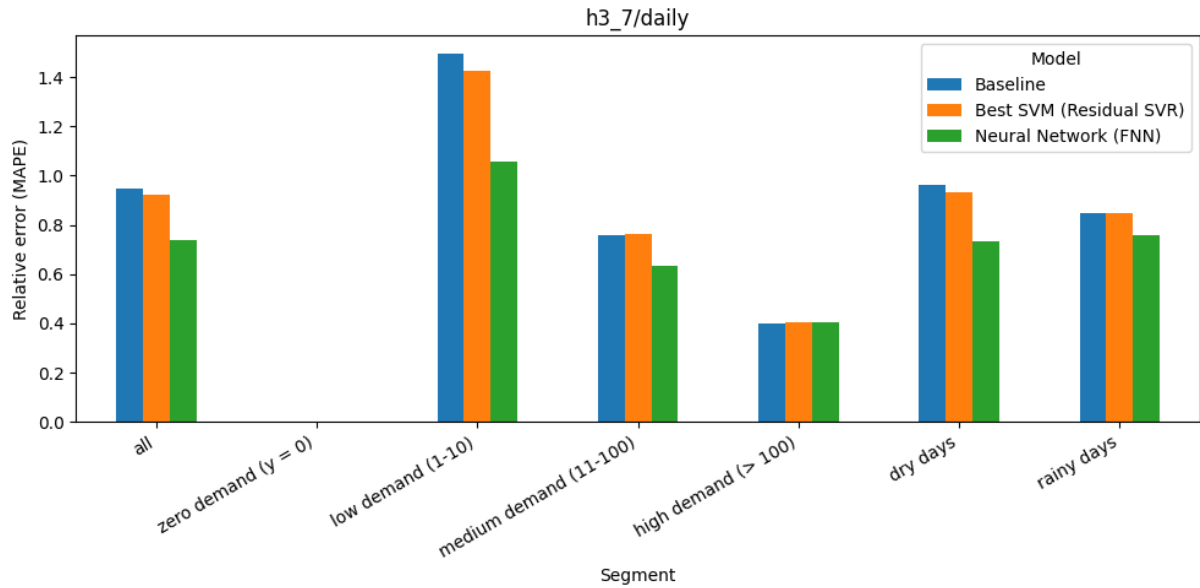
(a) h3_7/daily



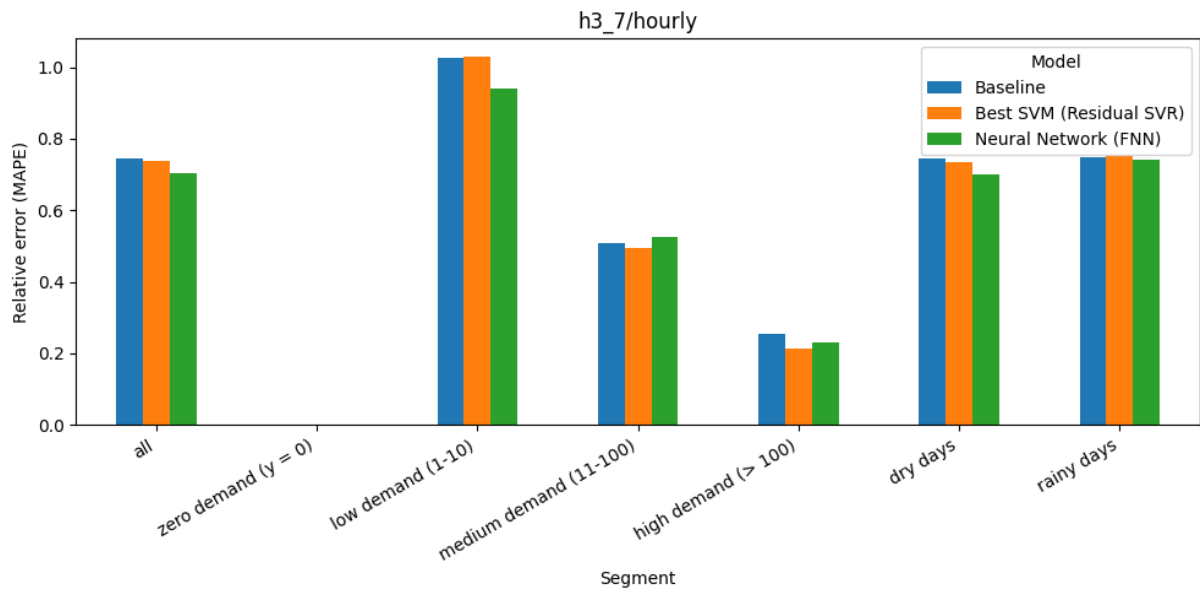
(b) h3_7/hourly

Figure 19: City-wide demand over the test period per temporal resolution.

--- h3_7/hourly ---



(a) h3_7/daily



(b) h3_7/hourly

Figure 20: Test relative error (MAE / mean demand) by segment per temporal resolution.

A.13 Reinforcement Learning (4.1)

No additional labeled figures or tables from this notebook.

References

Yan, J., Xiang, L., Wu, C., & Wu, H. (2020). City-scale taxi demand prediction using multisource urban geospatial data. *The International Archives of the Photogrammetry, Remote Sensing and Spatial Information Sciences*, 43, 213–220.

Table 7: Test MAE, relative error, and skill score by demand band and weather per temporal resolution.

(a) h3_7/daily		n	MAE	rel_error	skill_score
segment	Model				
all	Baseline	27972	21.391	0.948	0.000
	Best SVM (Residual SVR)	27972	20.639	0.923	0.035
	Neural Network (FNN)	27972	20.152	0.736	0.058
zero demand ($y = 0$)	Baseline	22149	0.126	NaN	0.000
	Best SVM (Residual SVR)	22149	0.189	NaN	-0.499
	Neural Network (FNN)	22149	0.188	NaN	-0.490
low demand (1-10)	Baseline	2384	4.149	1.494	0.000
	Best SVM (Residual SVR)	2384	3.974	1.427	0.042
	Neural Network (FNN)	2384	2.990	1.058	0.279
medium demand (11-100)	Baseline	1628	27.275	0.758	0.000
	Best SVM (Residual SVR)	1628	27.656	0.761	-0.014
	Neural Network (FNN)	1628	24.402	0.633	0.105
high demand (> 100)	Baseline	1811	298.878	0.401	0.000
	Best SVM (Residual SVR)	1811	286.379	0.404	0.042
	Neural Network (FNN)	1811	283.091	0.404	0.053
dry days	Baseline	24732	21.422	0.961	0.000
	Best SVM (Residual SVR)	24732	20.616	0.933	0.038
	Neural Network (FNN)	24732	19.679	0.733	0.081
rainy days	Baseline	3240	21.160	0.847	0.000
	Best SVM (Residual SVR)	3240	20.814	0.848	0.016
	Neural Network (FNN)	3240	23.766	0.760	-0.123
(b) h3_7/hourly		n	MAE	rel_error	skill_score
segment	Model				
all	Baseline	671328	1.068	0.746	0.000
	Best SVM (Residual SVR)	671328	1.016	0.738	0.048
	Neural Network (FNN)	671328	0.977	0.704	0.085
zero demand ($y = 0$)	Baseline	620830	0.043	NaN	0.000
	Best SVM (Residual SVR)	620830	0.107	NaN	-1.482
	Neural Network (FNN)	620830	0.031	NaN	0.277
low demand (1-10)	Baseline	26513	3.324	1.026	0.000
	Best SVM (Residual SVR)	26513	3.321	1.028	0.001
	Neural Network (FNN)	26513	3.171	0.939	0.046
medium demand (11-100)	Baseline	17335	15.334	0.507	0.000
	Best SVM (Residual SVR)	17335	14.823	0.494	0.033
	Neural Network (FNN)	17335	15.912	0.527	-0.038
high demand (> 100)	Baseline	6650	50.534	0.256	0.000
	Best SVM (Residual SVR)	6650	40.714	0.215	0.194
	Neural Network (FNN)	6650	41.552	0.231	0.178
dry days	Baseline	590544	1.058	0.746	0.000
	Best SVM (Residual SVR)	590544	1.004	0.736	0.051
	Neural Network (FNN)	590544	0.966	0.698	0.087
rainy days	Baseline	80784	1.143	0.749	0.000
	Best SVM (Residual SVR)	80784	1.110	0.752	0.028
	Neural Network (FNN)	80784	1.057	0.742	0.075